

Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-12 20:17:50

Data & configuration

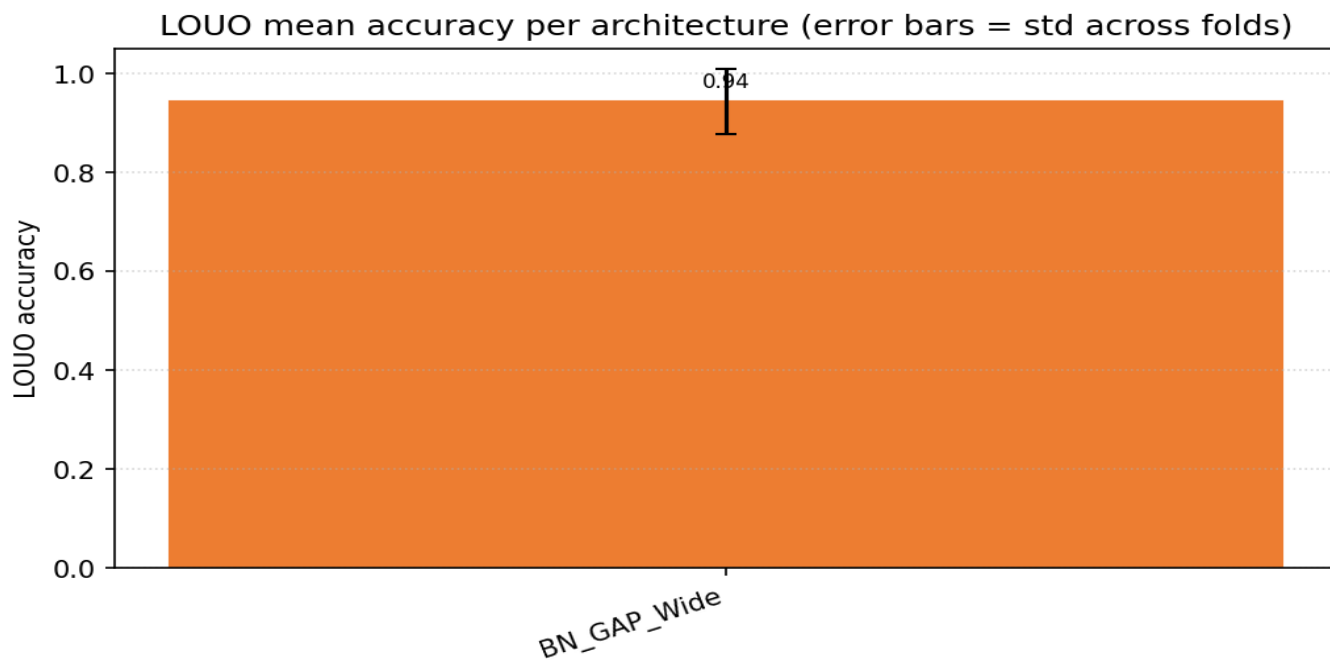
LOUO path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
LOUO path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
LOUO path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
LOUO path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
LOUO path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
LOUO path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
LOUO path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
LOUO path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
LOUO path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
LOUO path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
LOUO path 11	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic
LOUO path 12	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
LOUO path 13	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
LOUO users loaded	13 (11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Per-trial normalisation	zscore
Downstream normalisation	minmax
Sampling rate	30.0 Hz
Random state	42
Augmentation	on · copies=1 ·
Epochs · batch size	40 · 16
Early stopping	on · monitor val_loss · patience=8
LOUO inner-validation	user
LOUO variants	BN_GAP_Wide
Final-retrain variant	Deep
Final-retrain held-out user	— (trained on all users)

Variant comparison (LOUO)

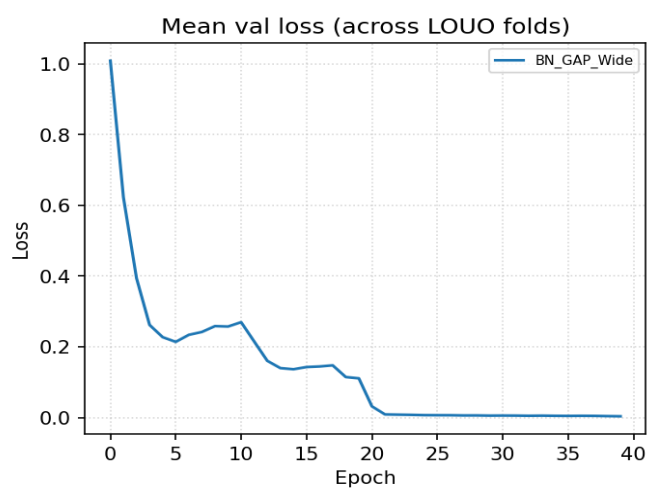
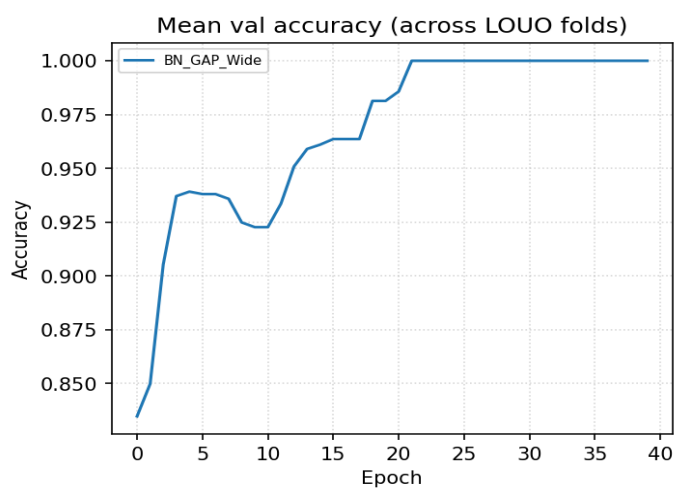
Variant	Params	LOUO mean	LOUO std	Folds	Held-out user acc
BN_GAP_Wide	104,423	0.9436	0.0656	13	—

Best LOUO mean: BN_GAP_Wide

LOUO accuracy per variant

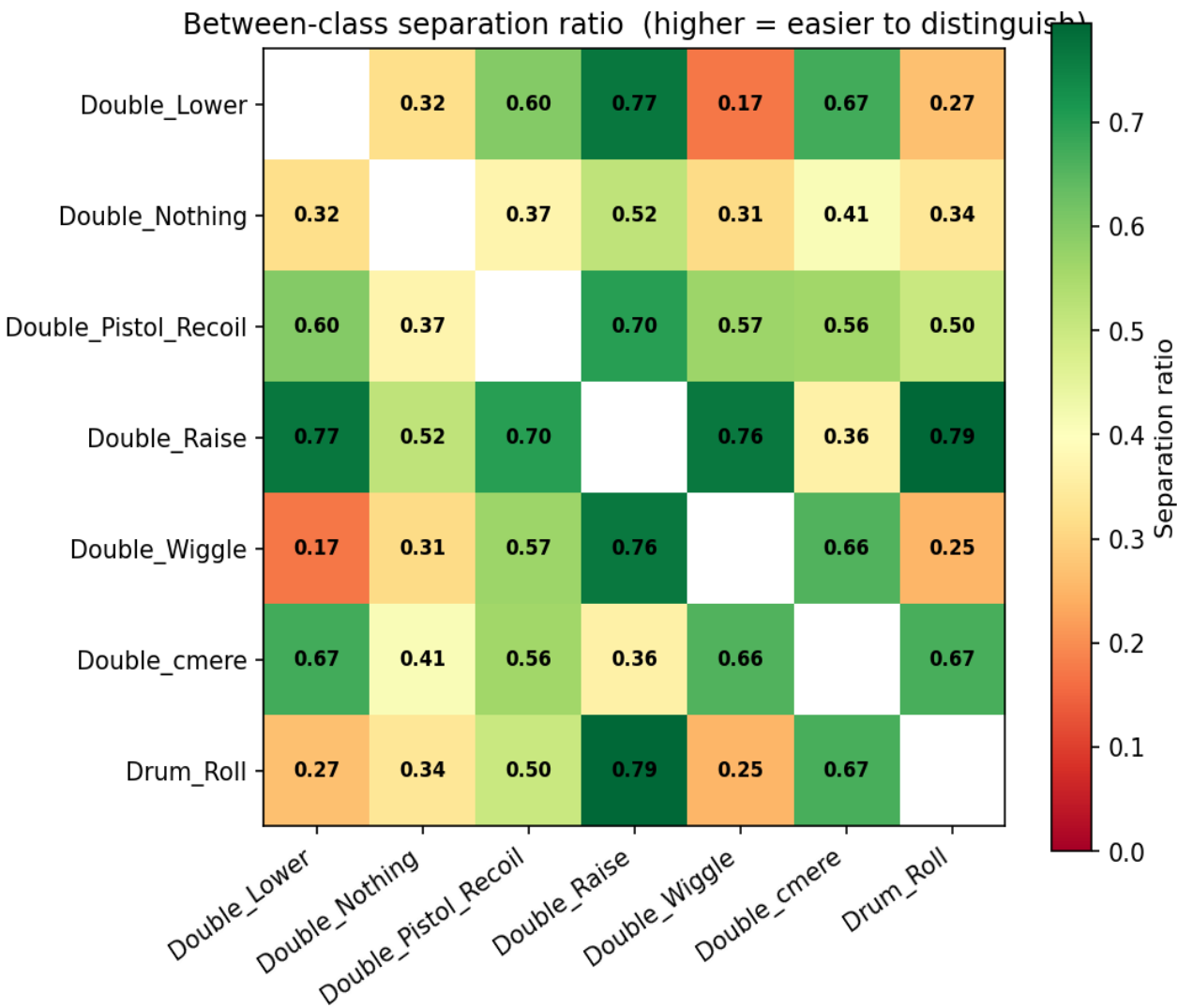
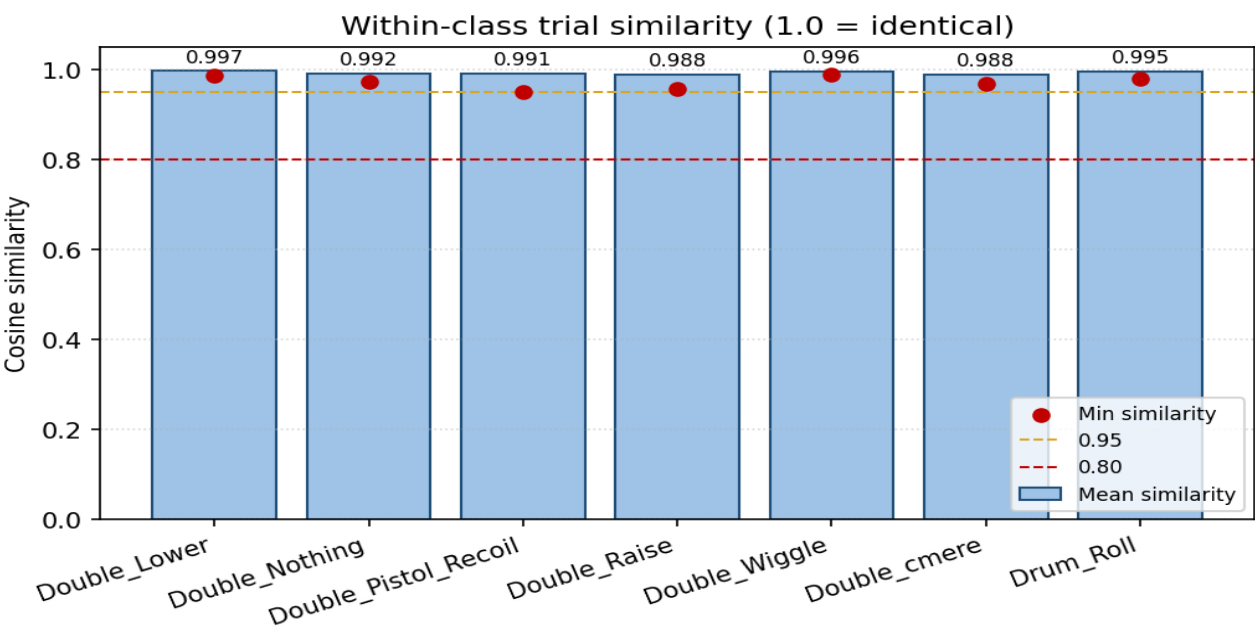


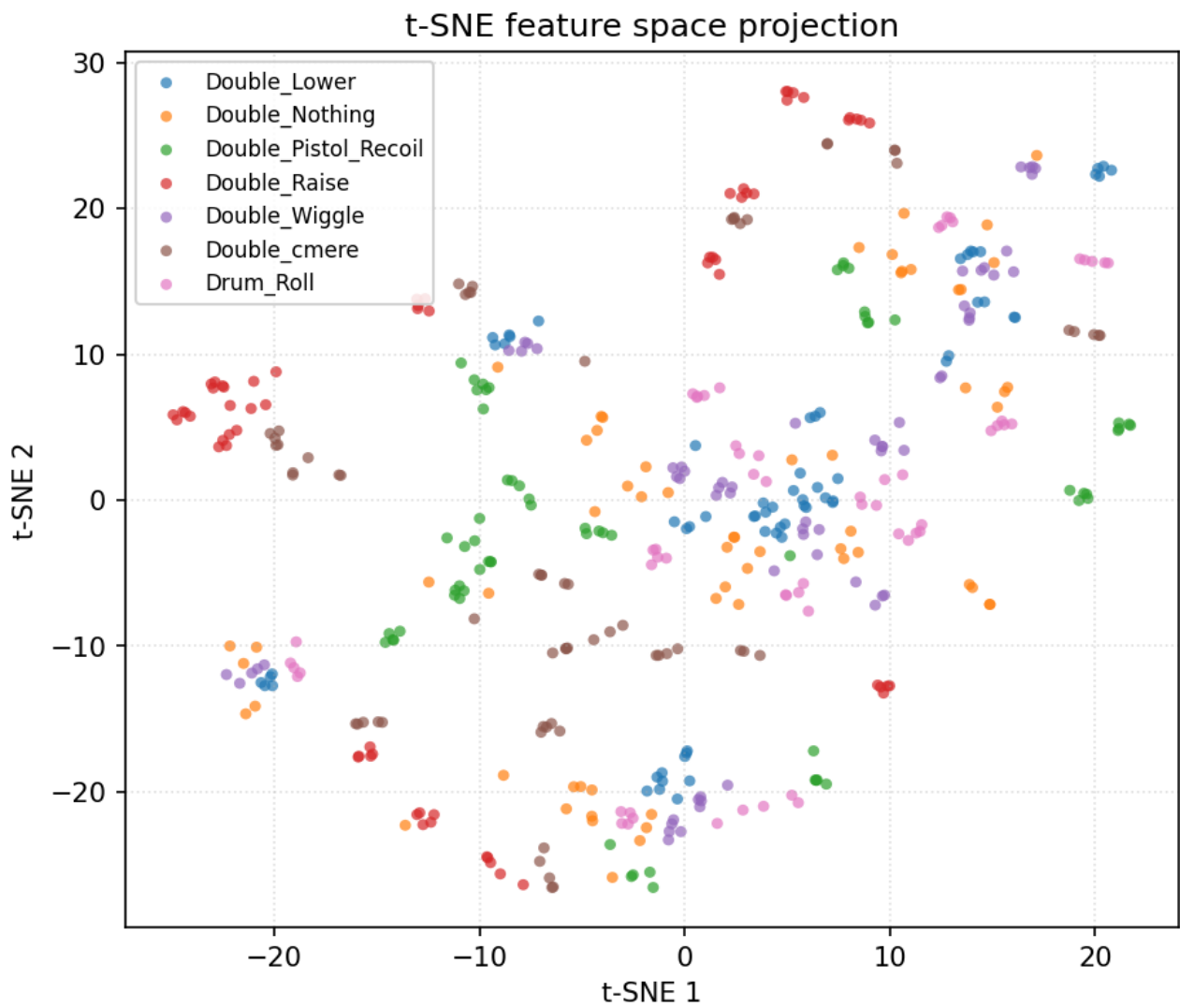
Mean training curves across LOUO folds



Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).

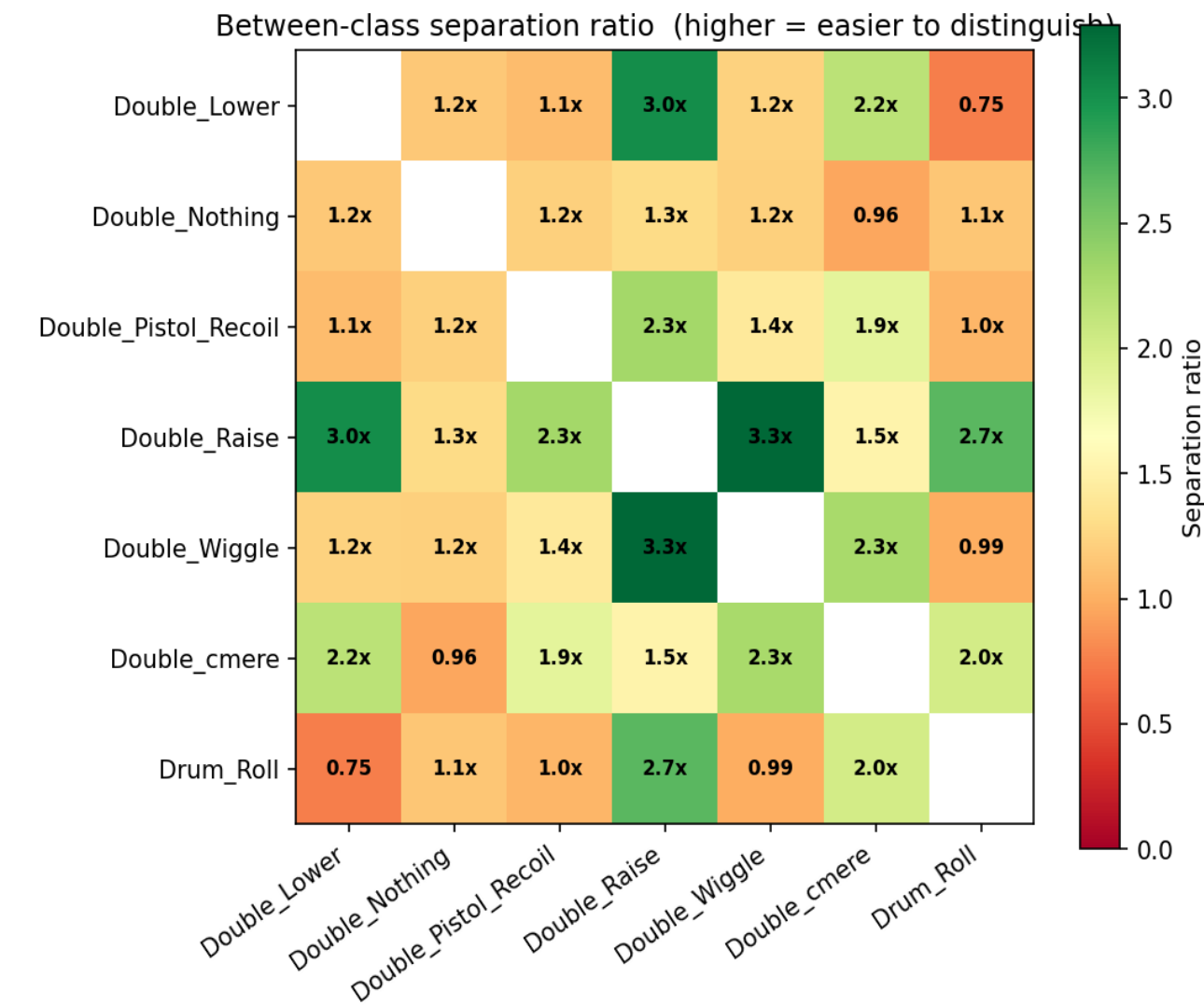
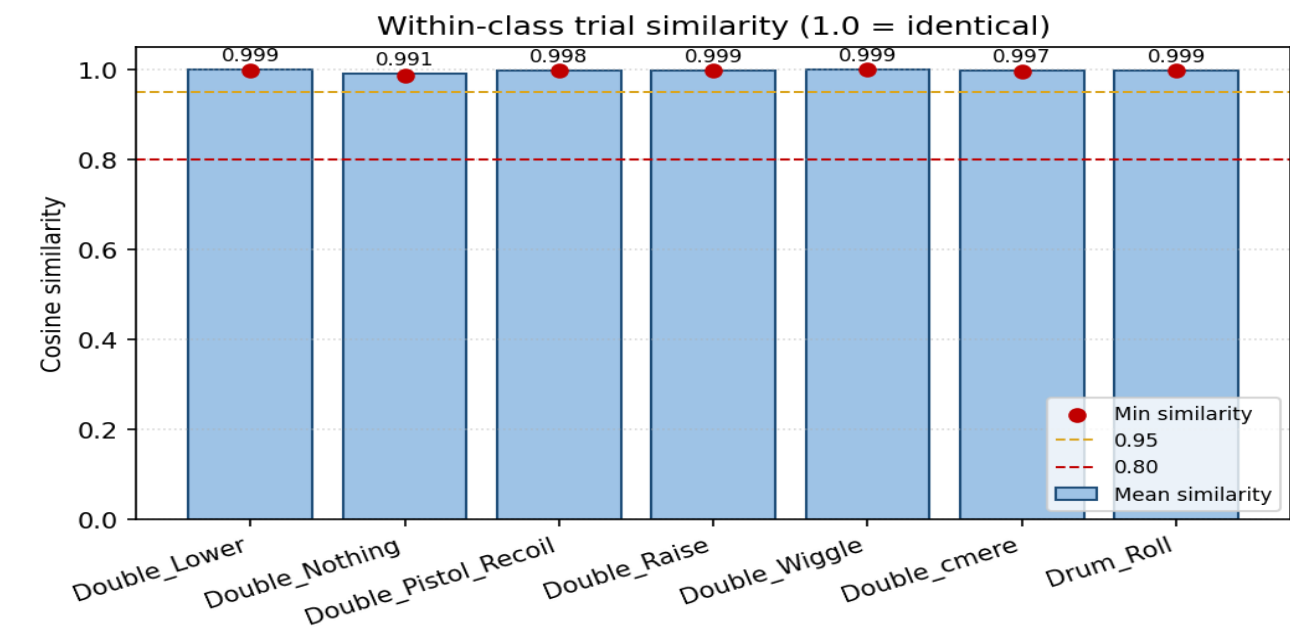




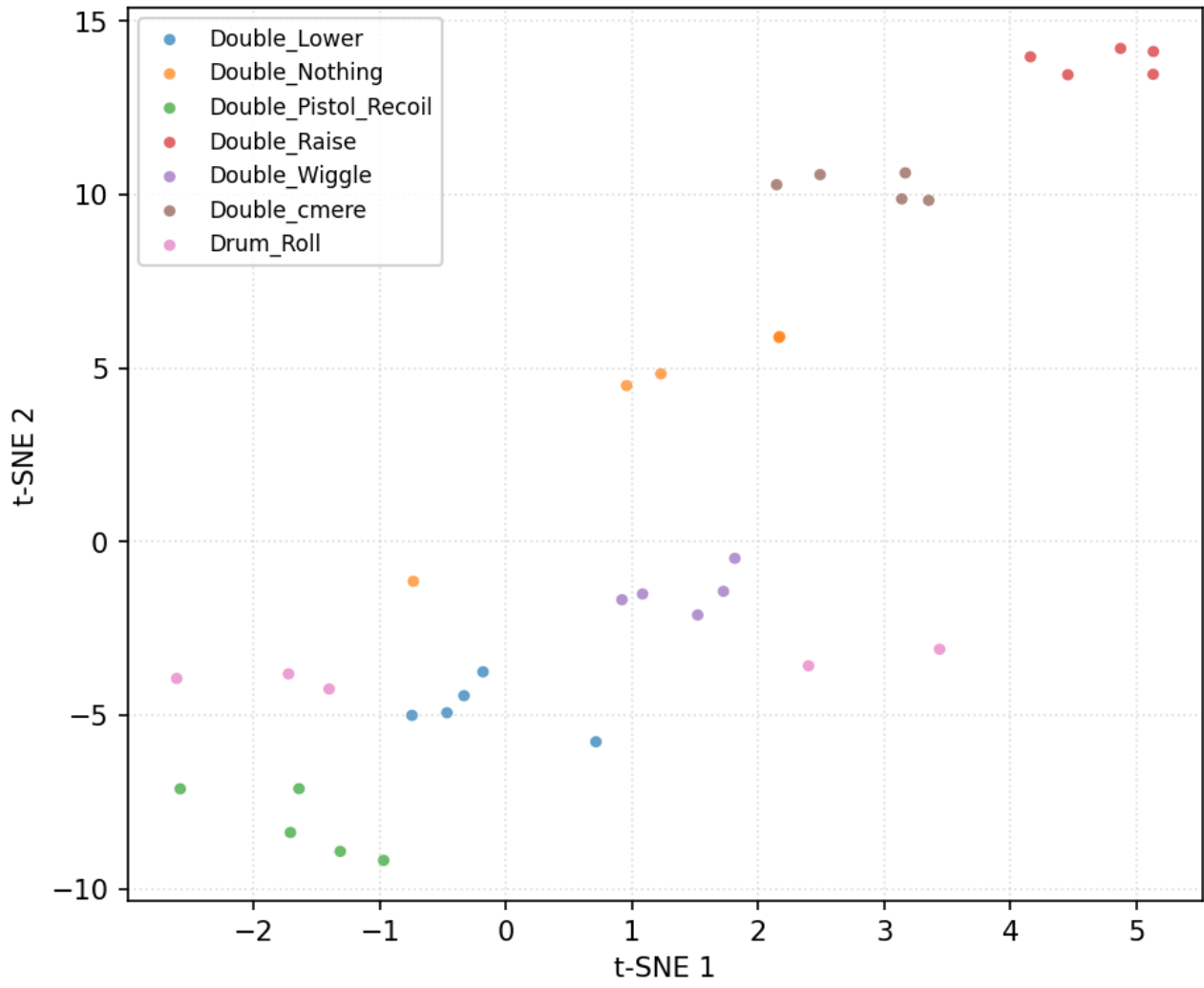
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

Data diagnostics — user 11_Mansh

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

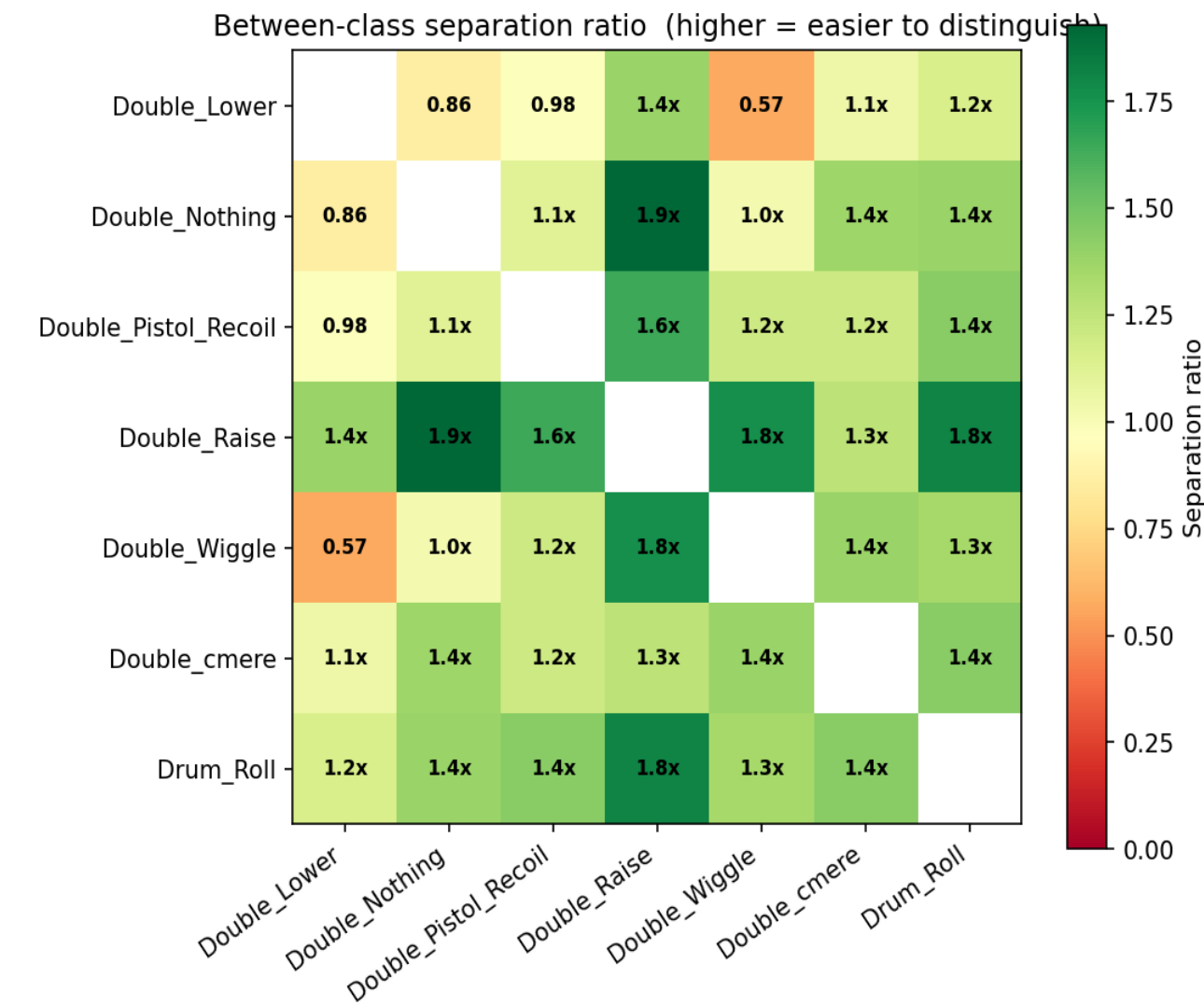
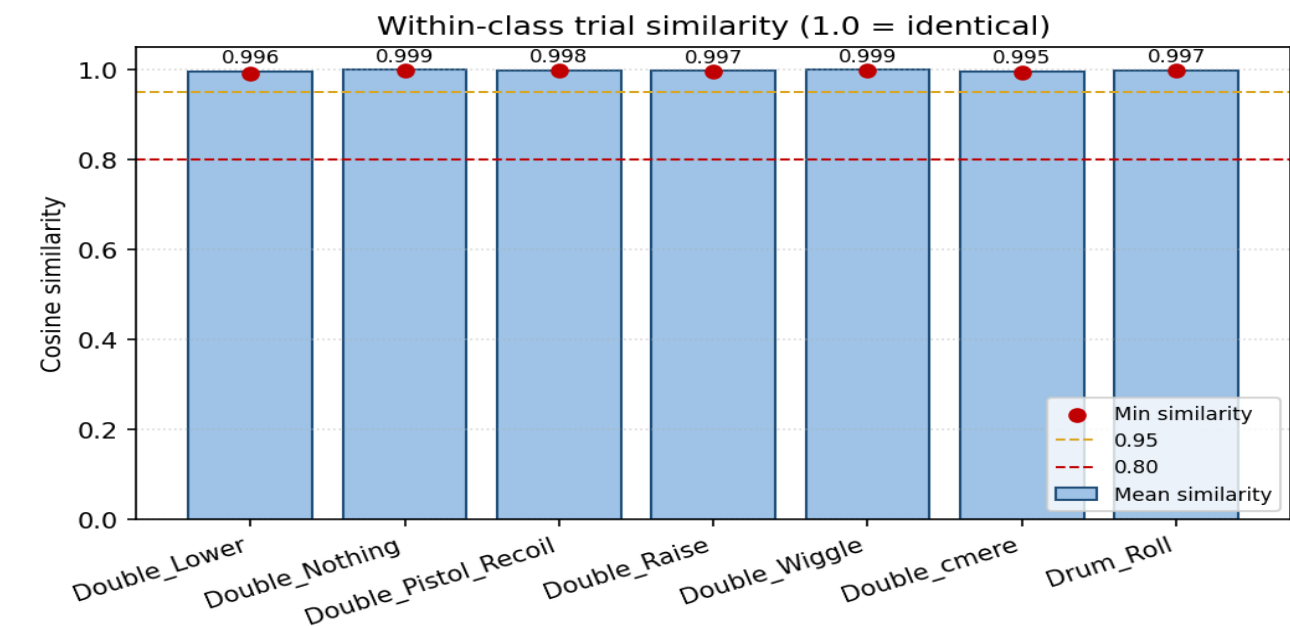


t-SNE feature space projection

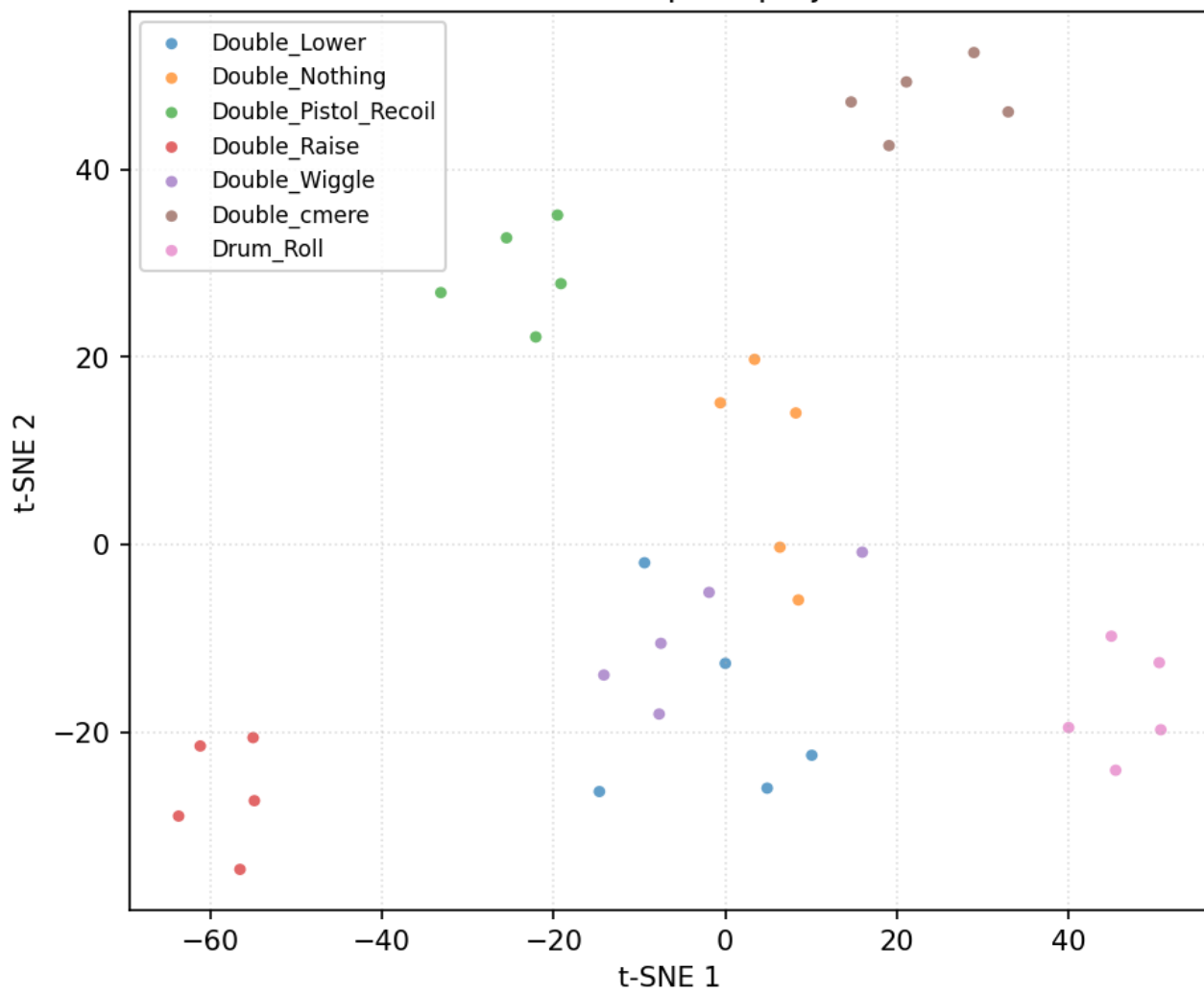


Data diagnostics — user 12_Marcus

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

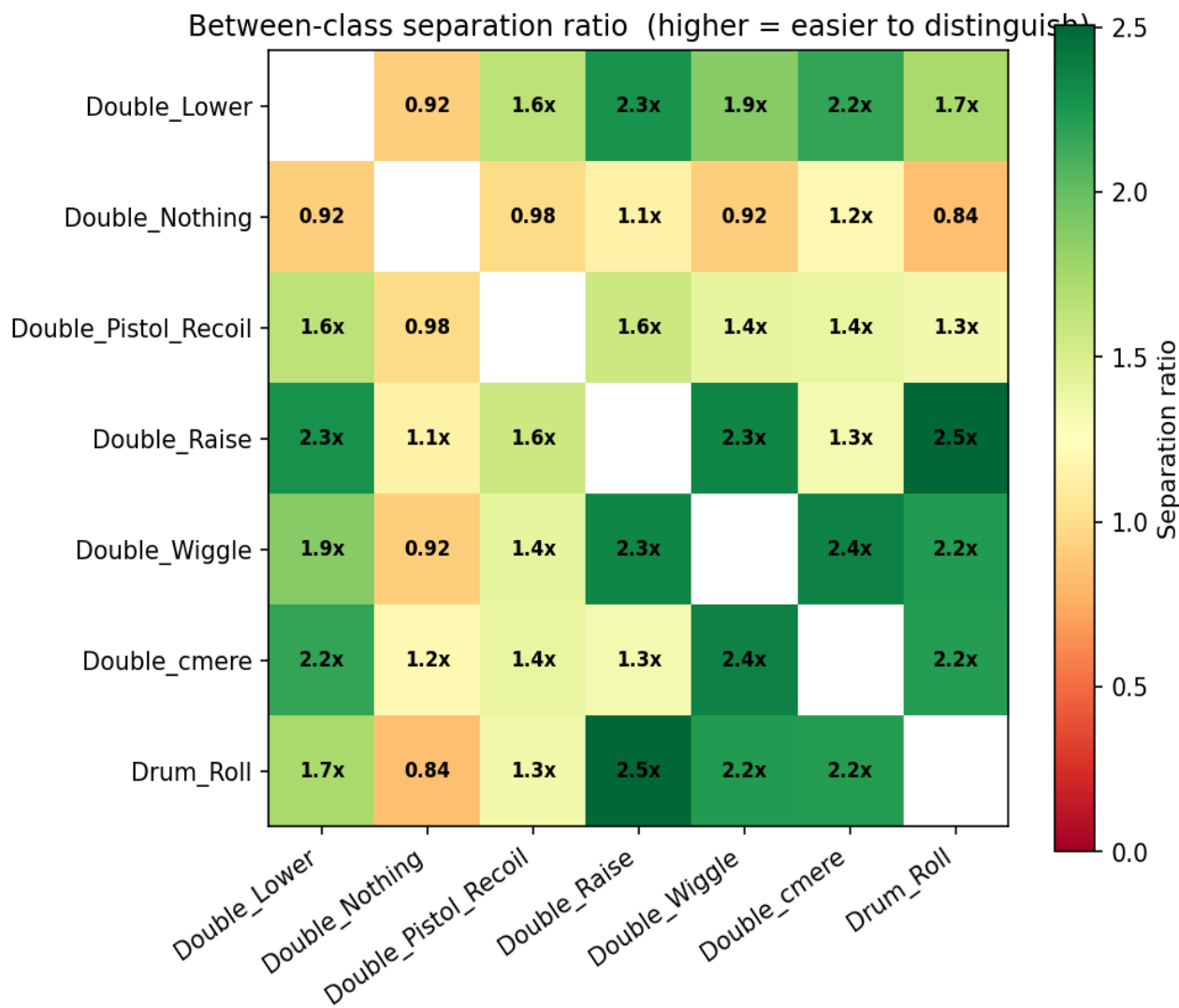
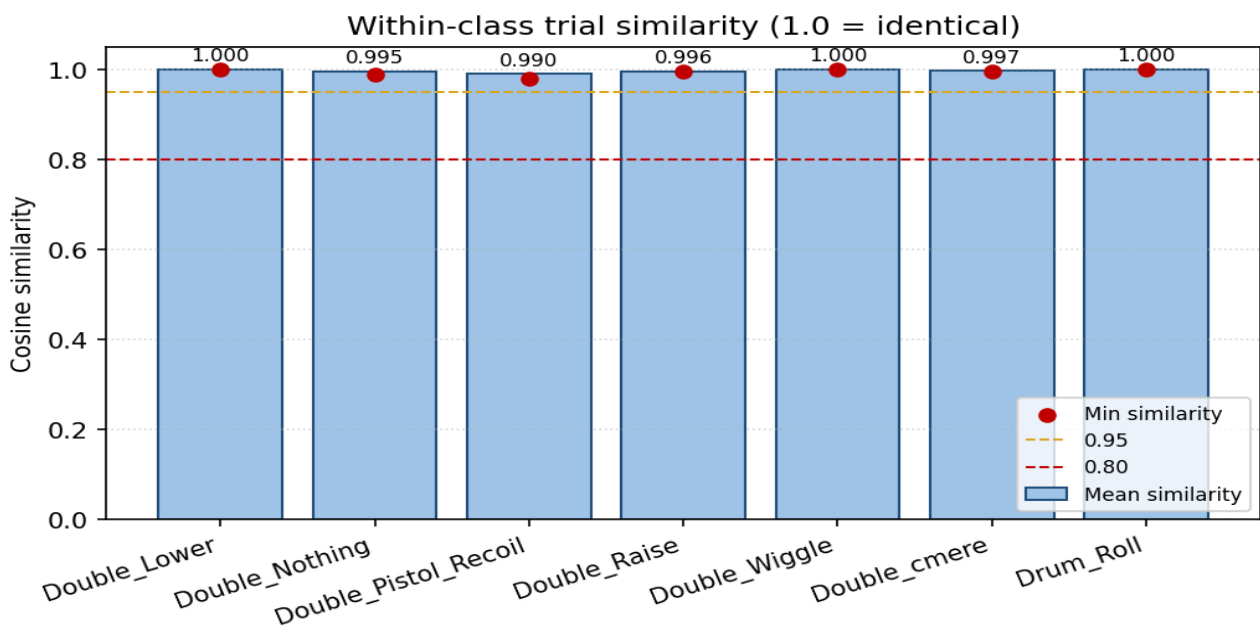


t-SNE feature space projection

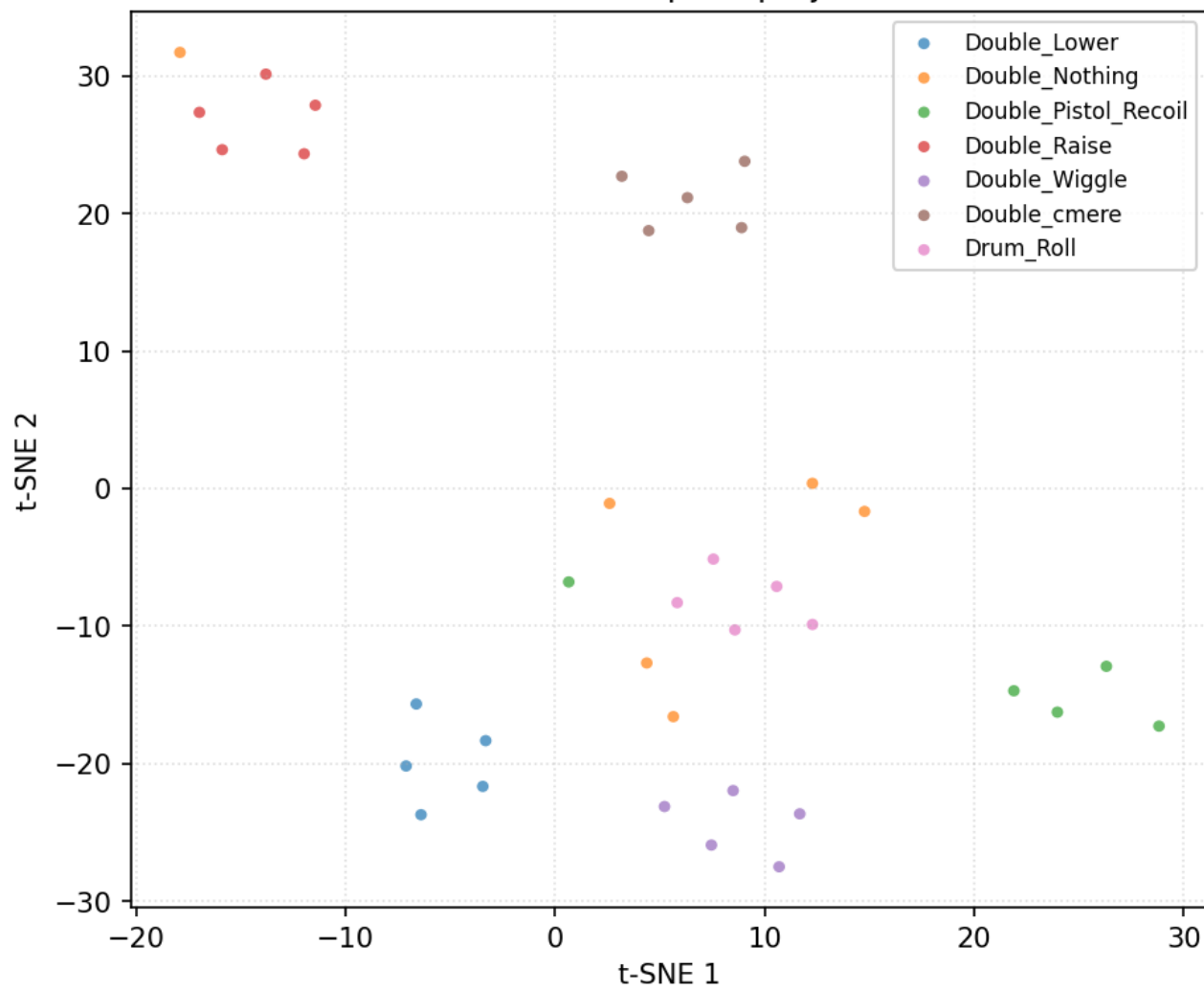


Data diagnostics — user 14_StephenV2

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

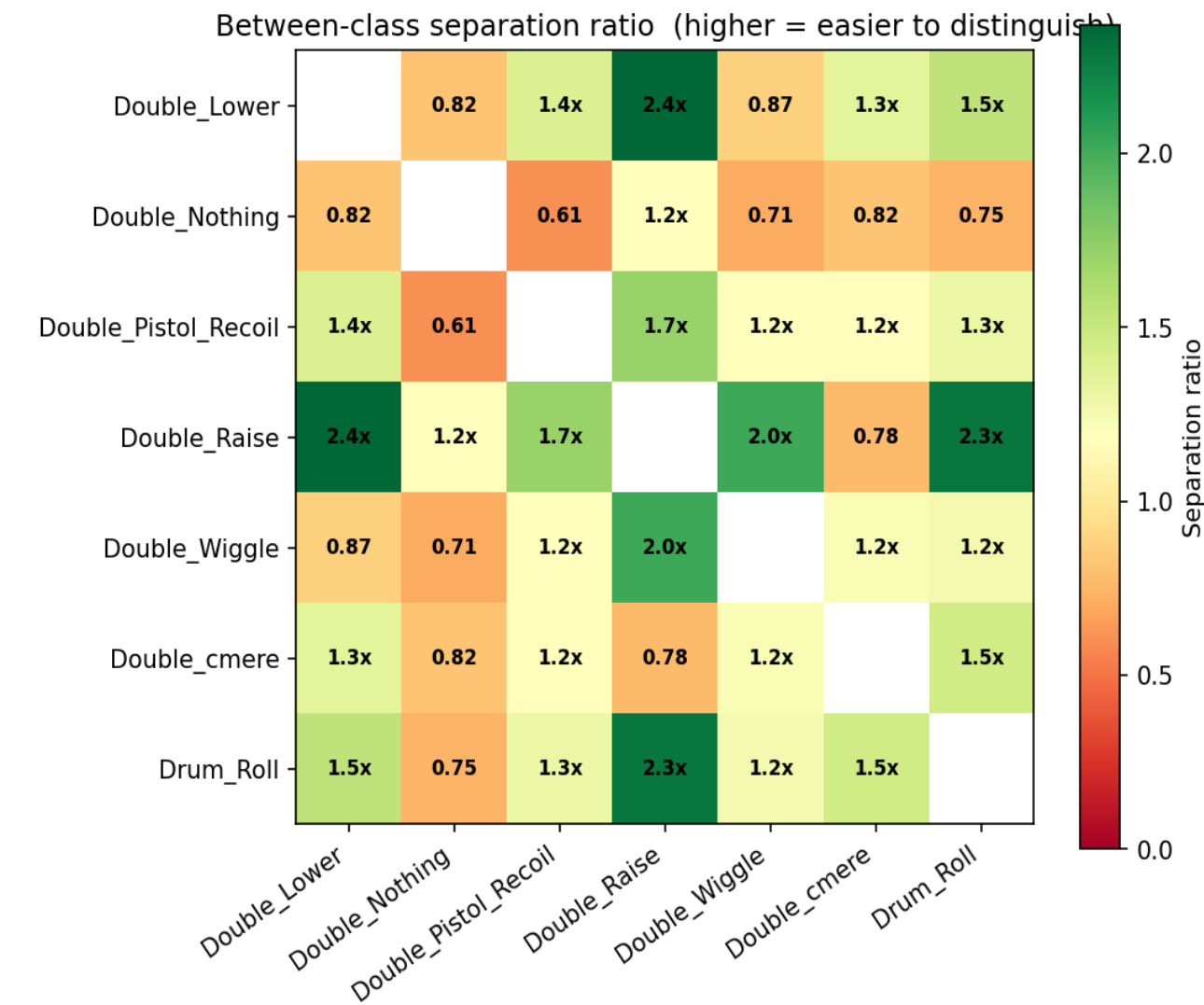
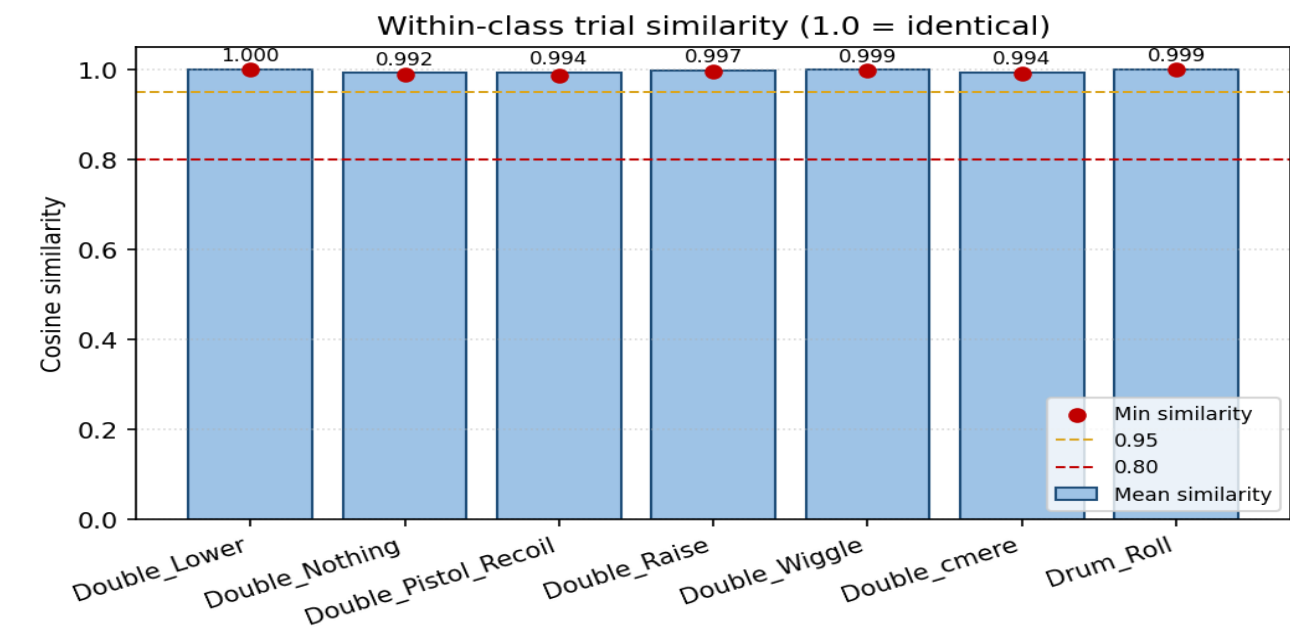


t-SNE feature space projection

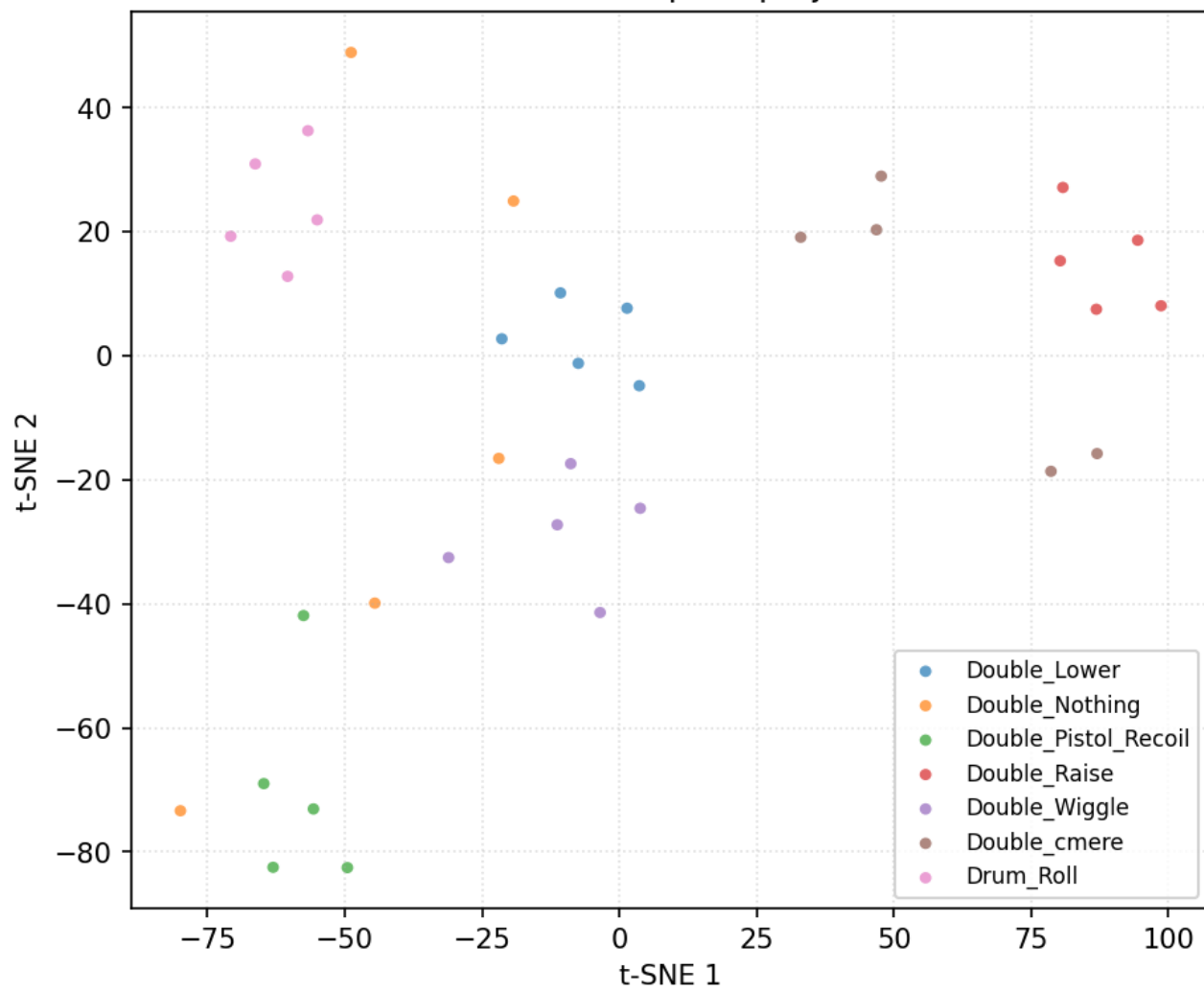


Data diagnostics — user 15_Tash

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

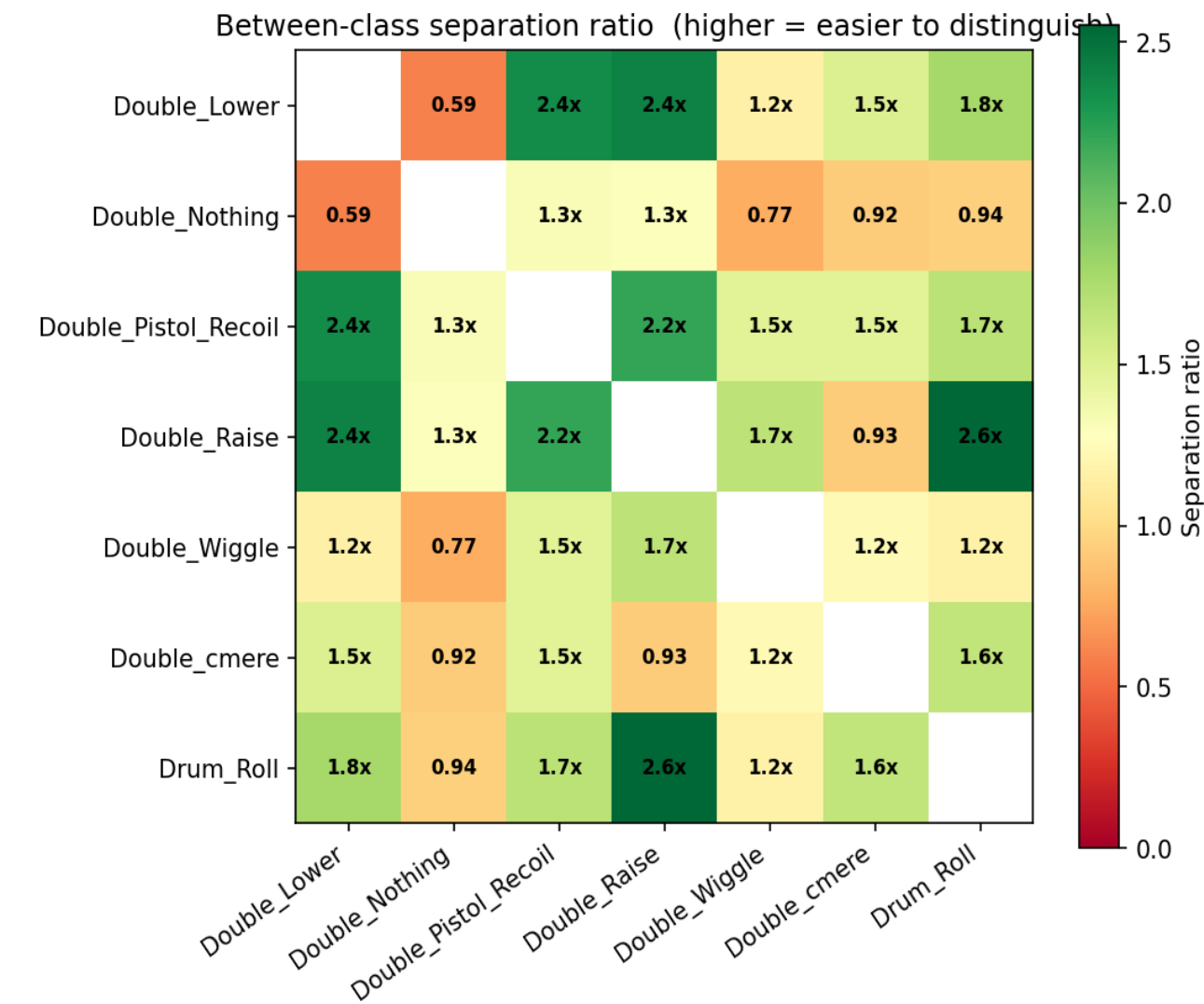
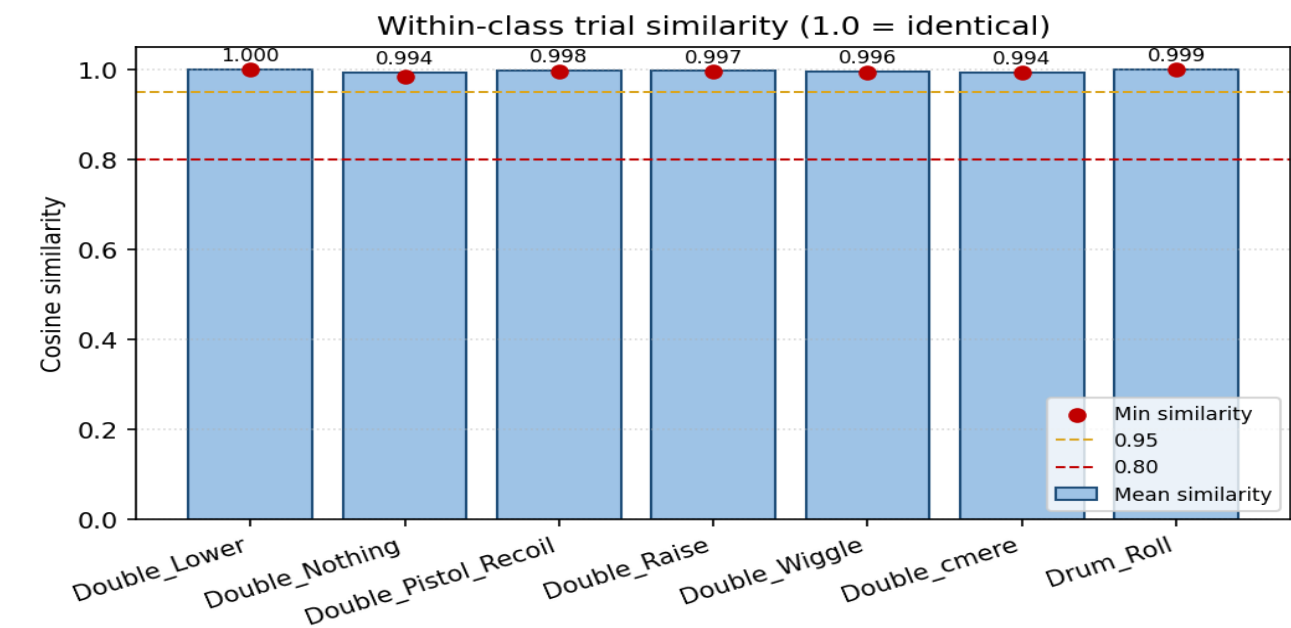


t-SNE feature space projection

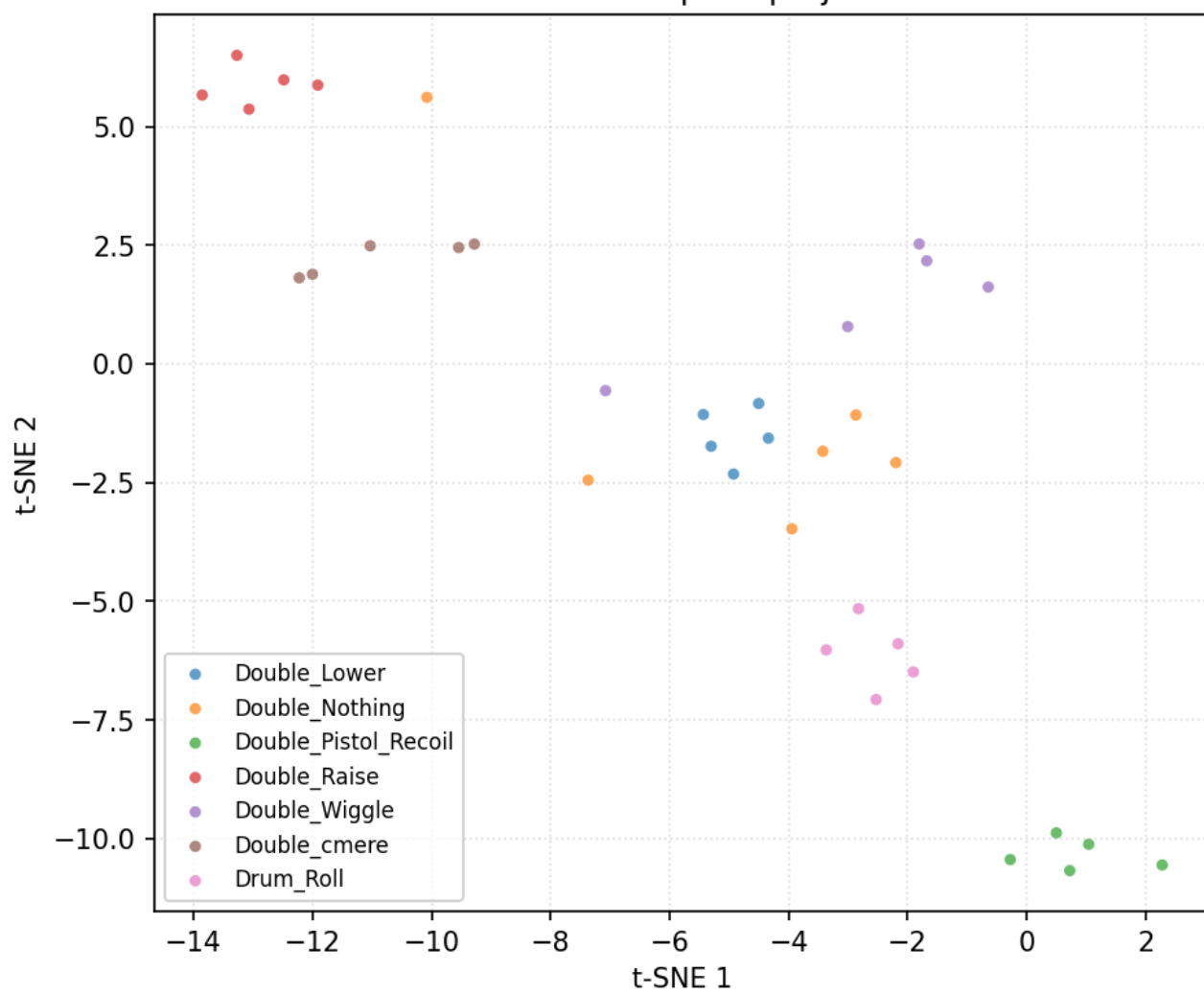


Data diagnostics — user 16_Bella

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

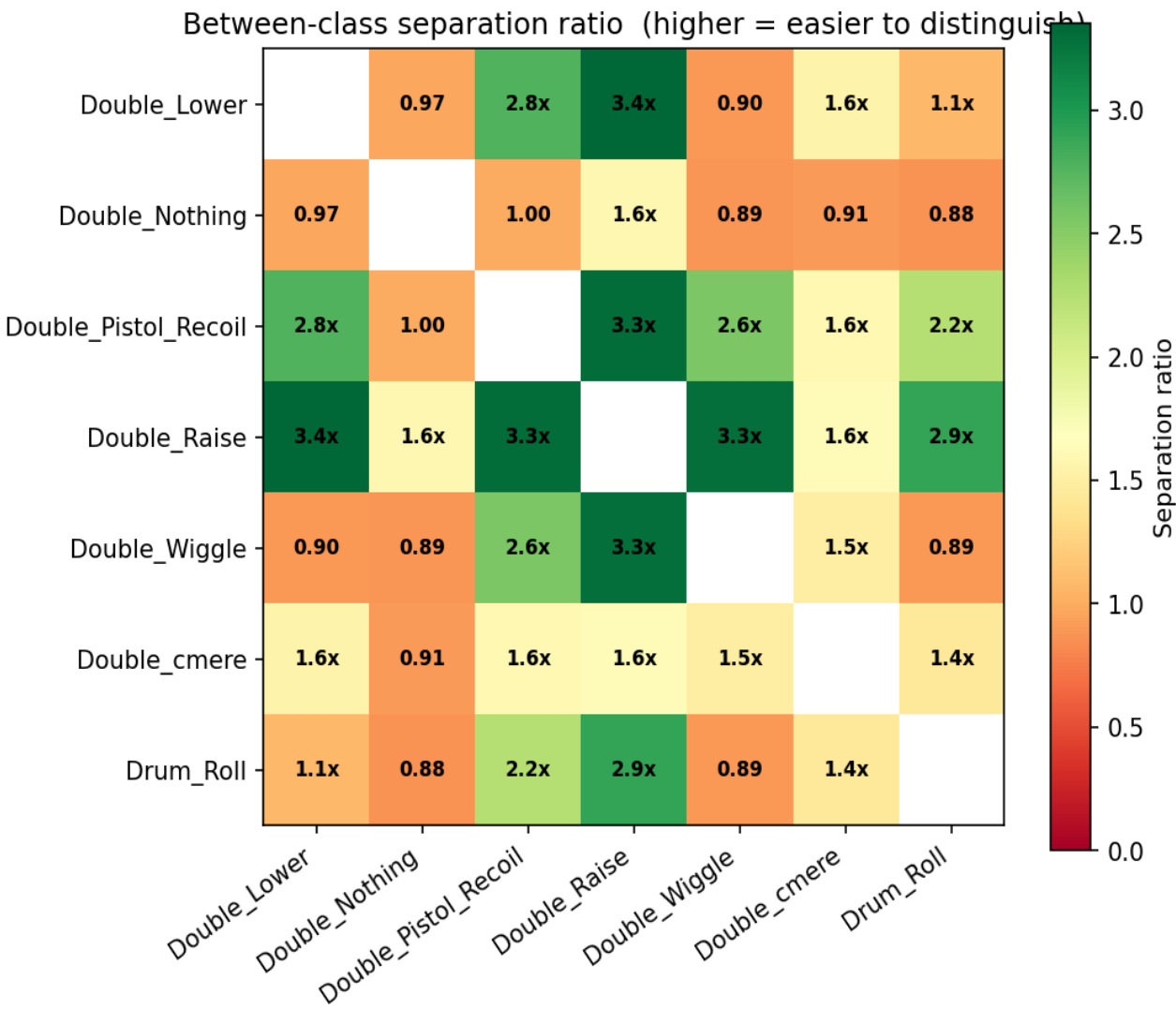
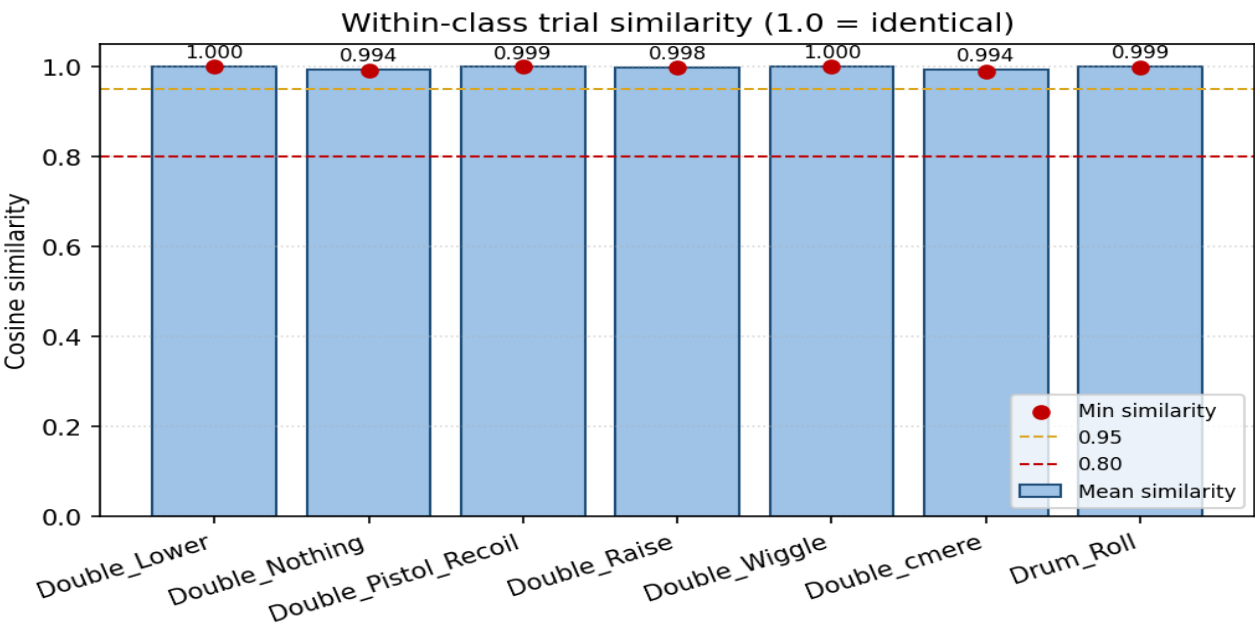


t-SNE feature space projection

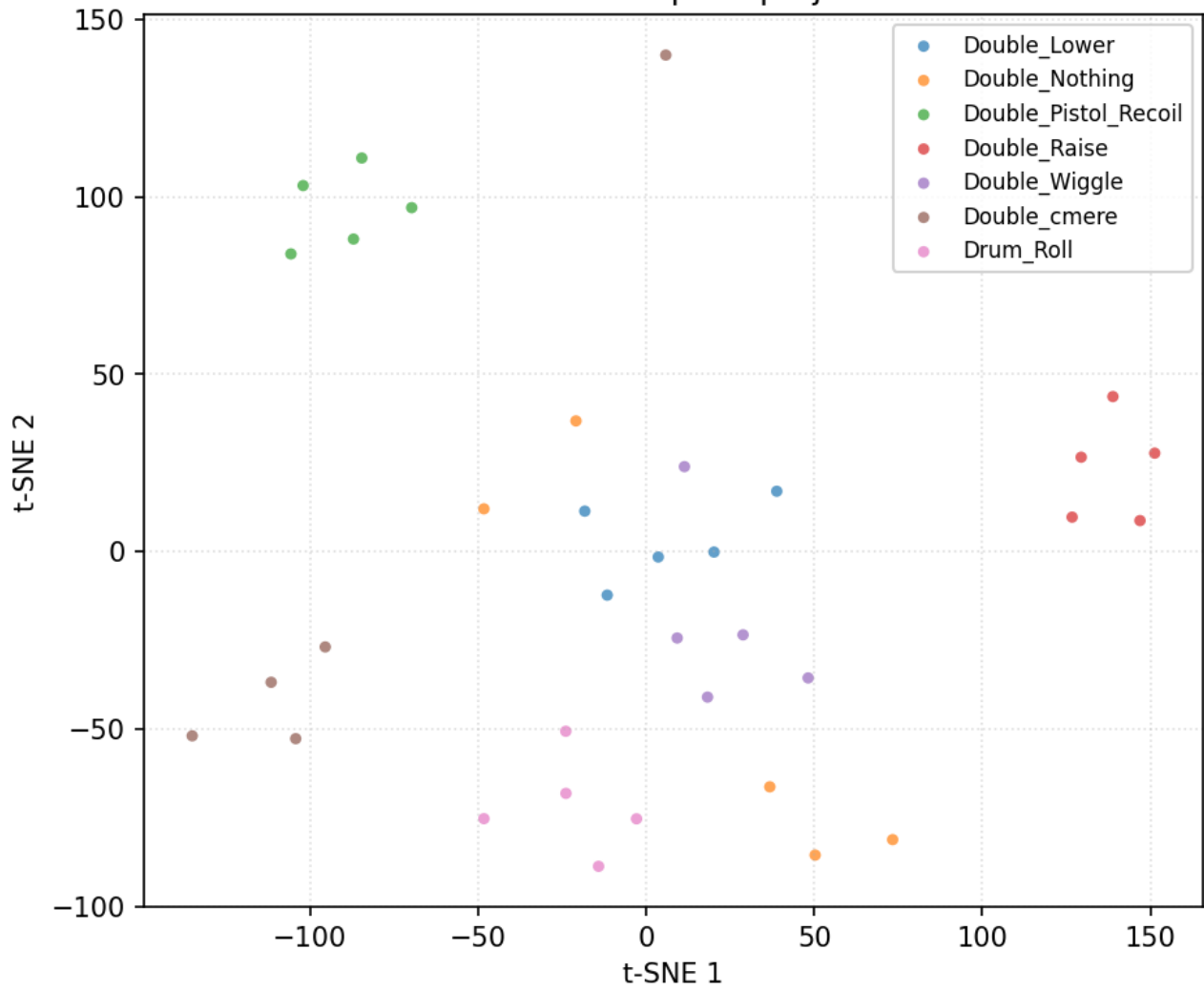


Data diagnostics — user 17_Raquel

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

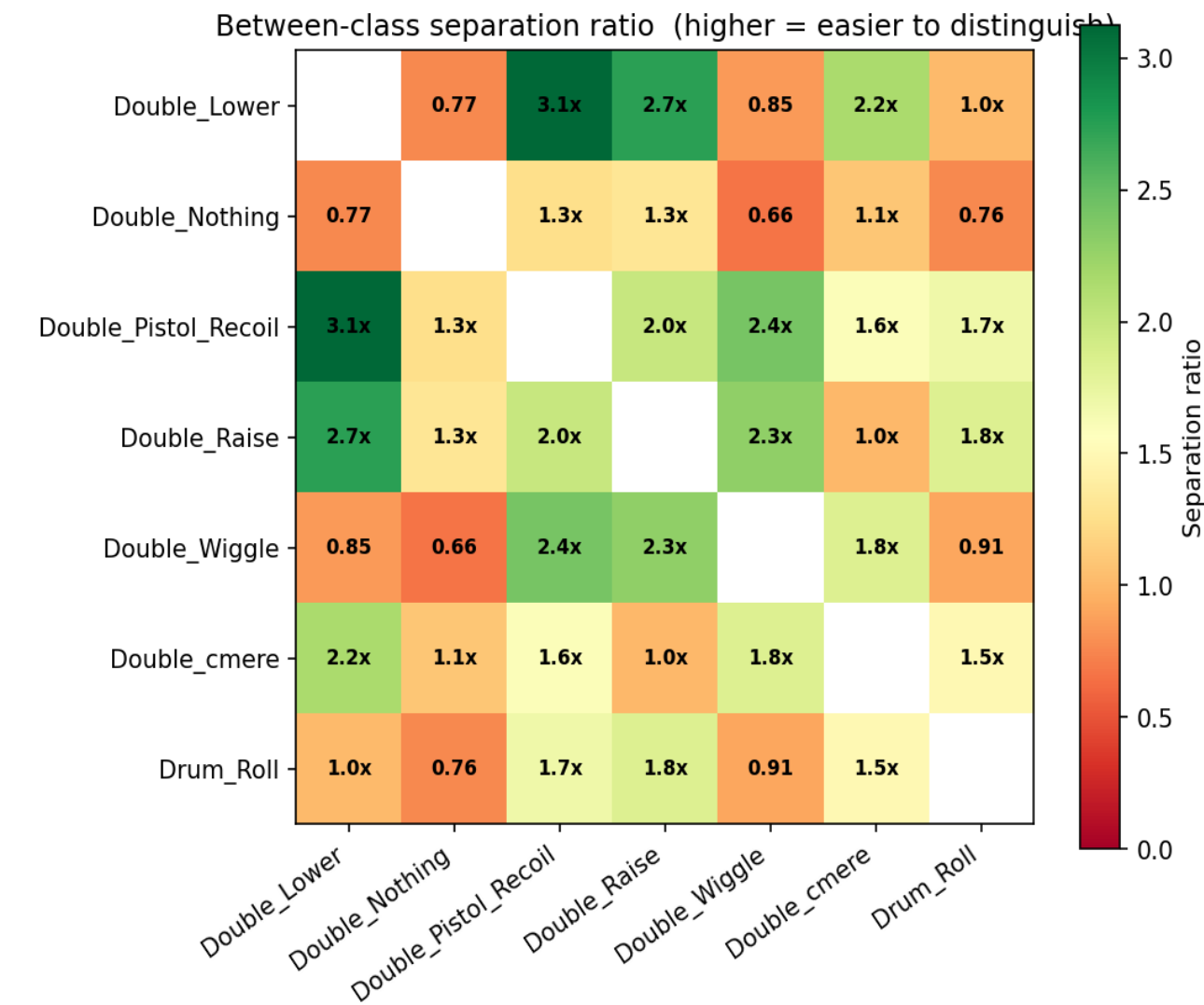
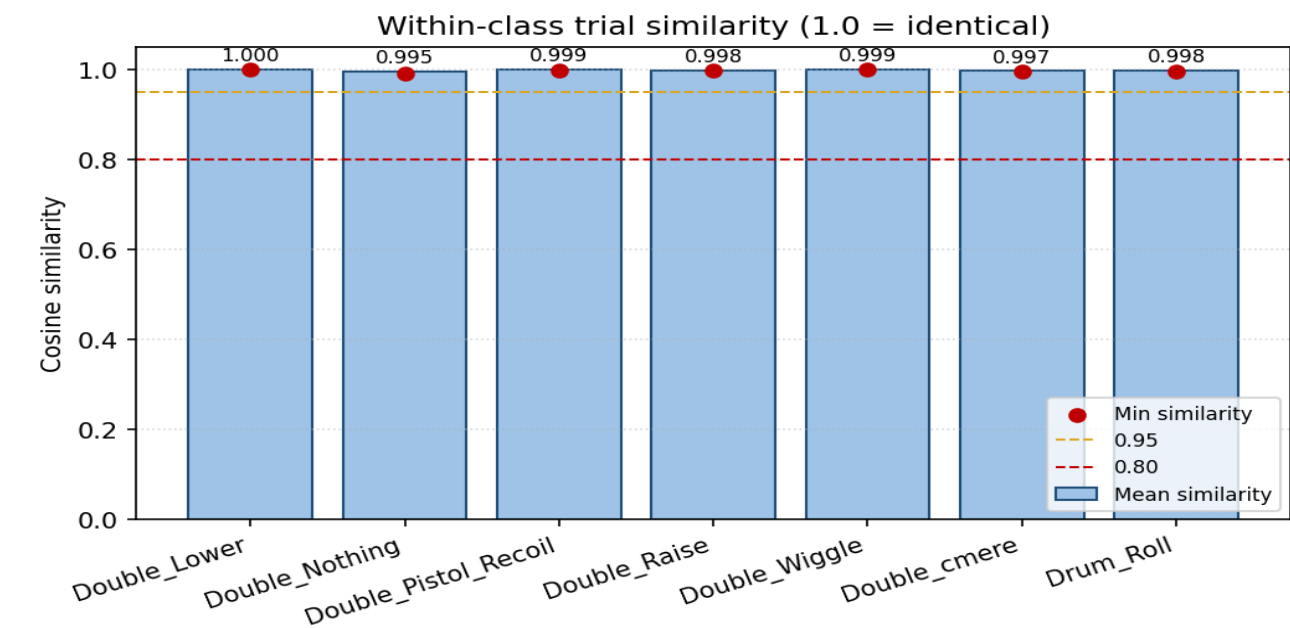


t-SNE feature space projection

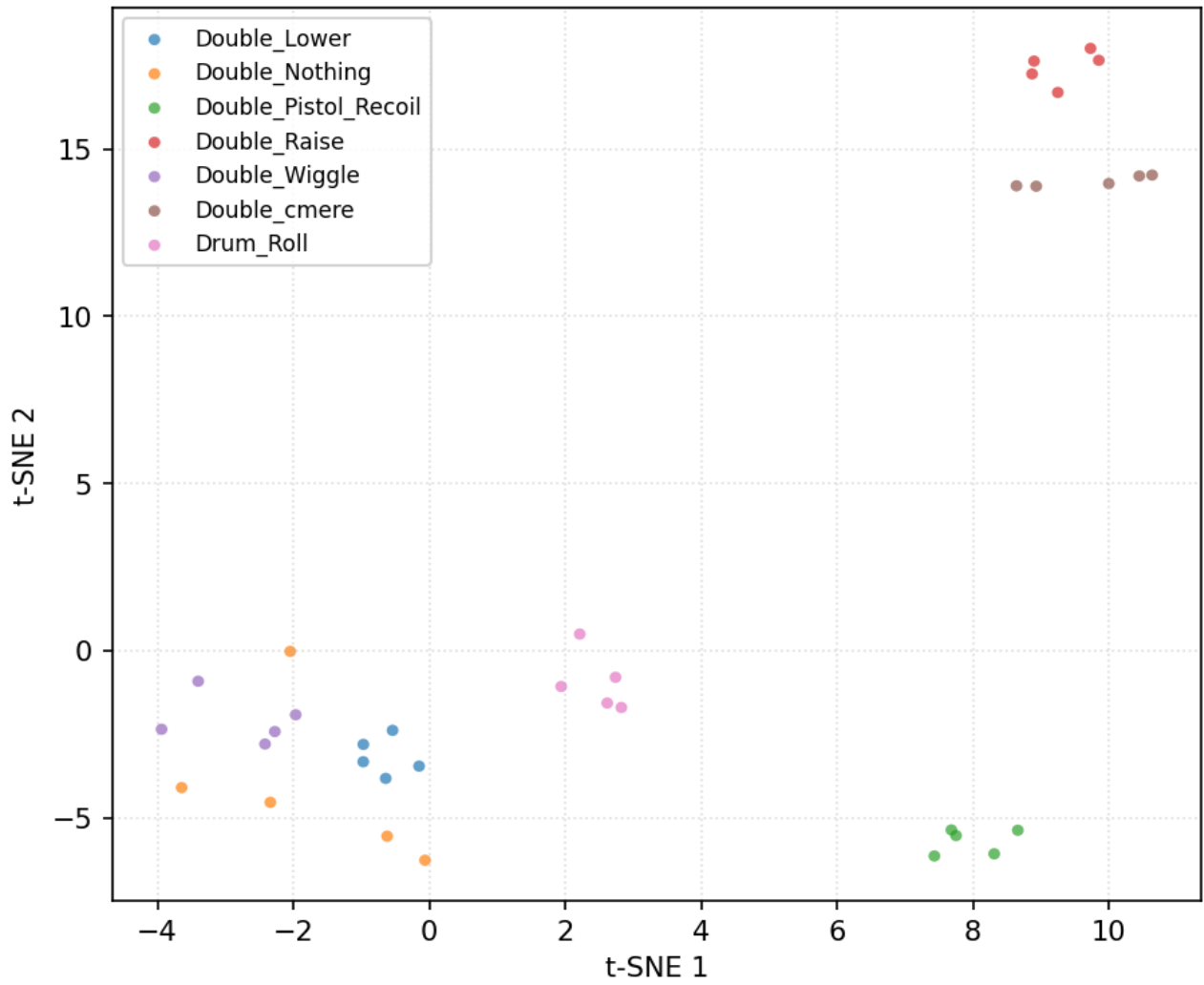


Data diagnostics — user 18_Bridgette

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

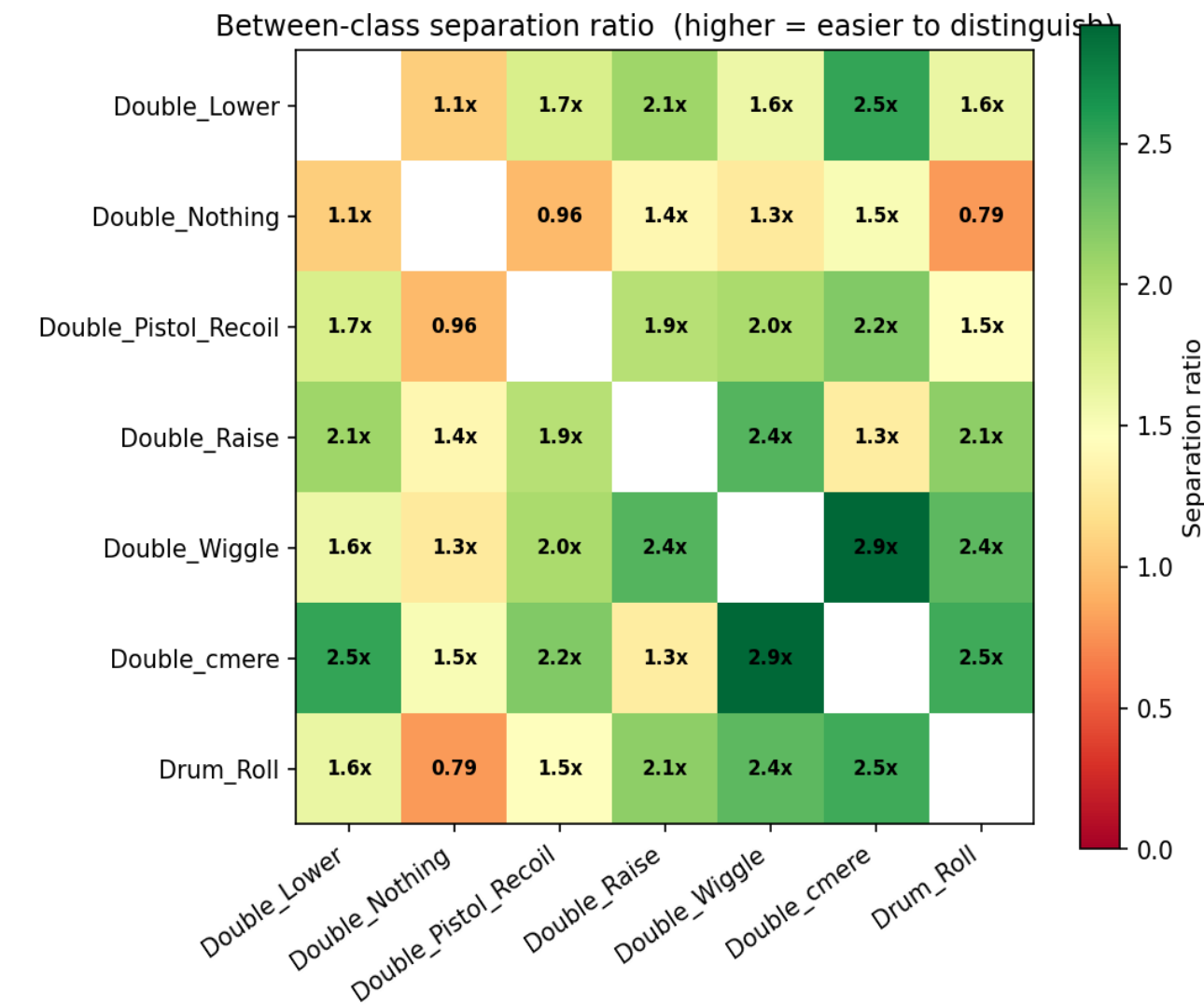
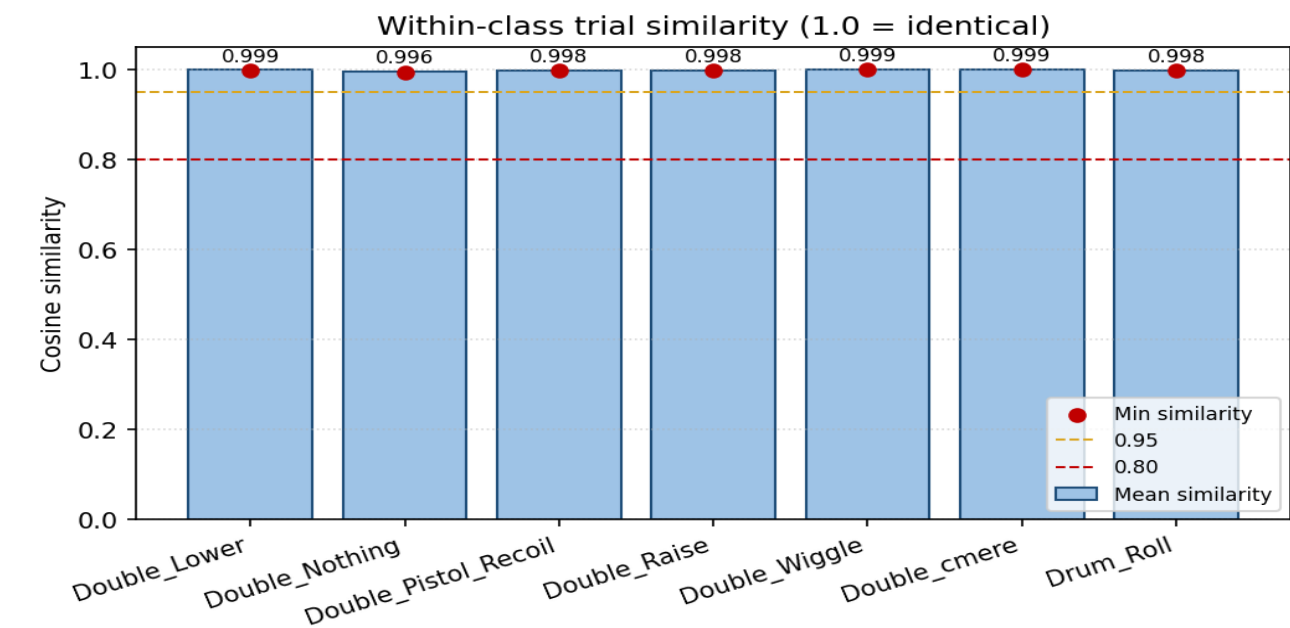


t-SNE feature space projection

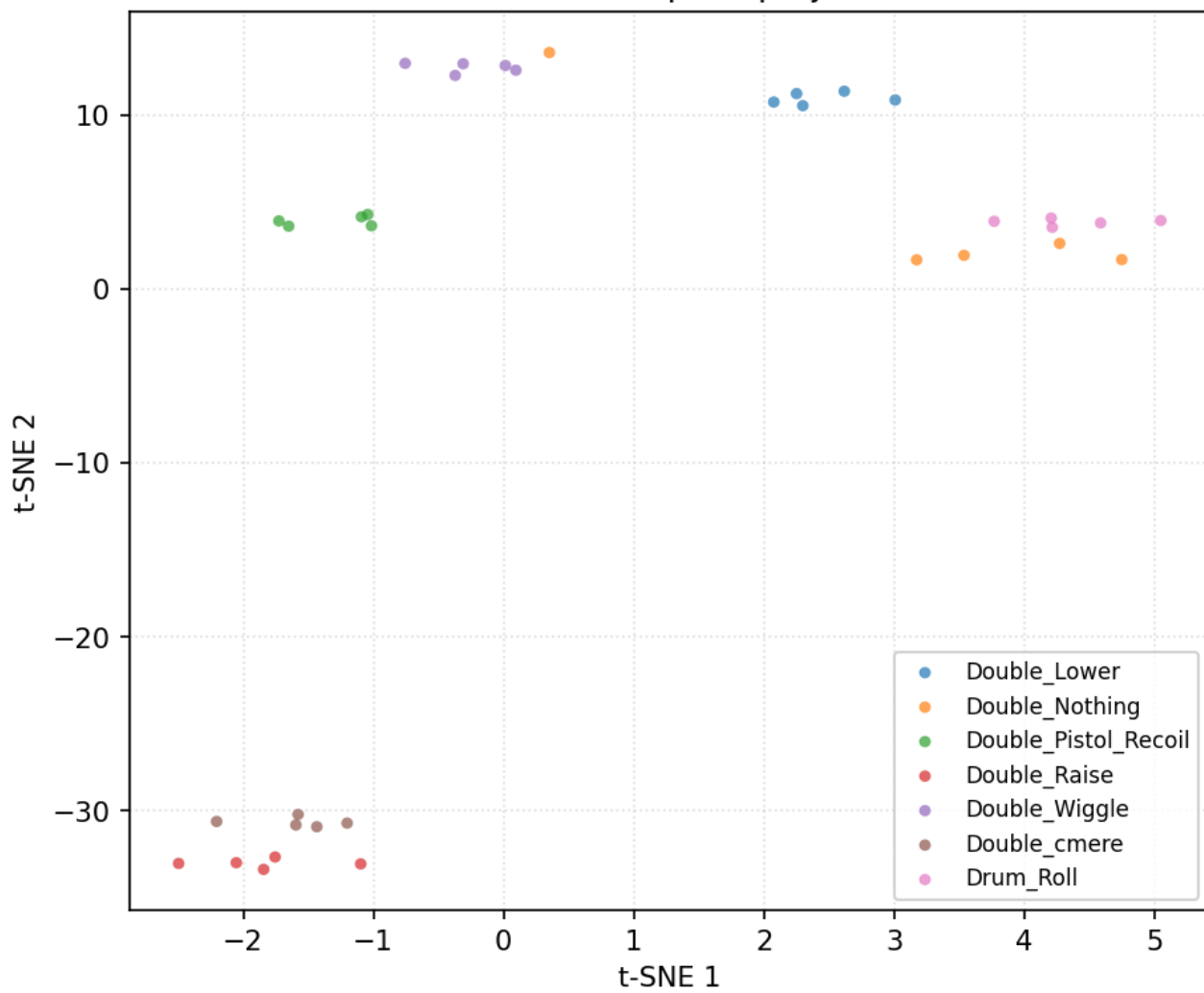


Data diagnostics — user 1_Alán

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

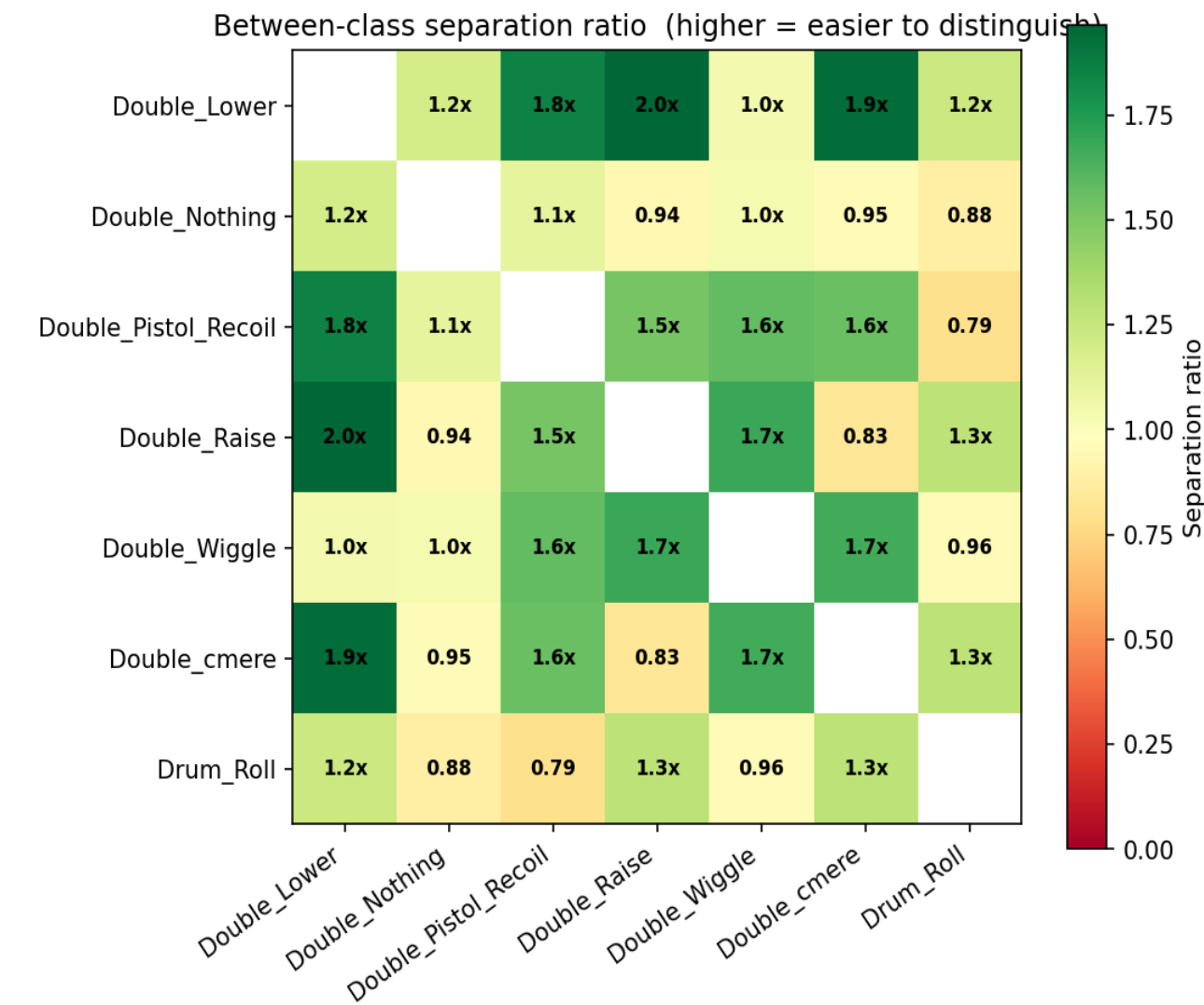
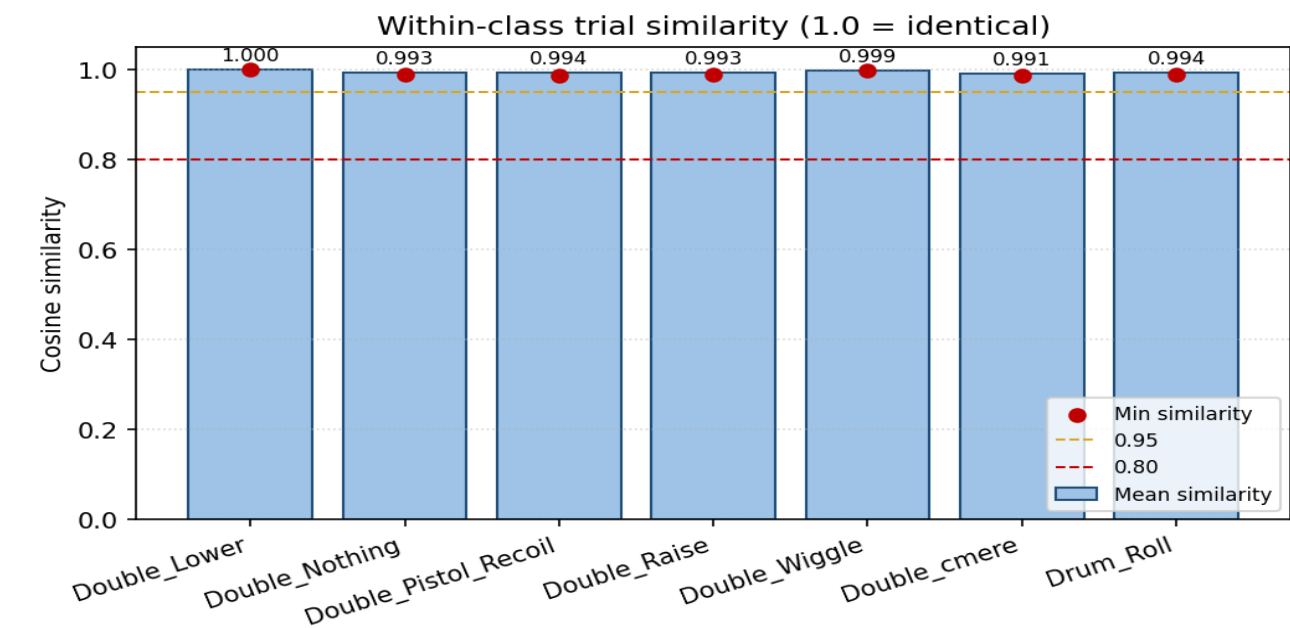


t-SNE feature space projection

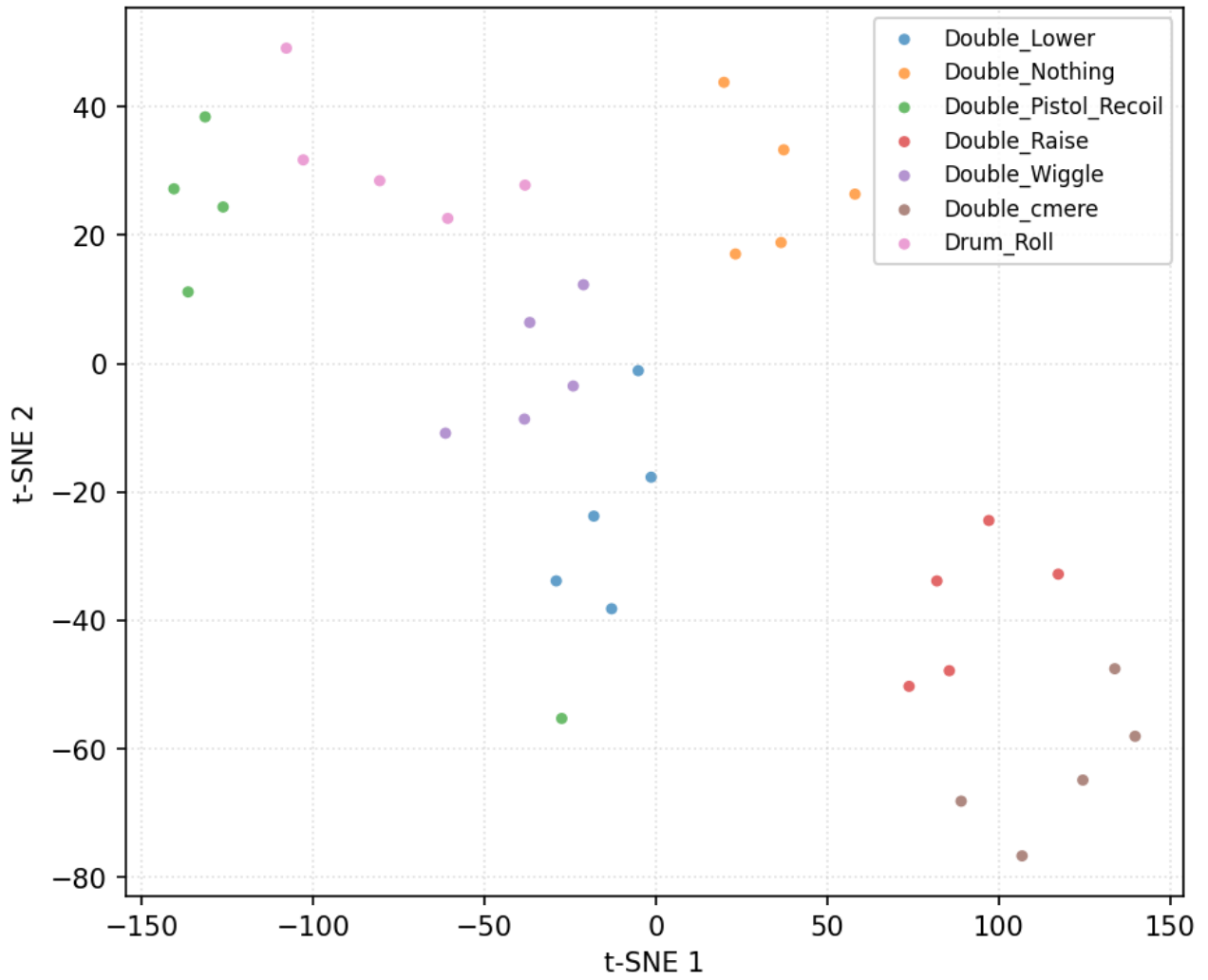


Data diagnostics — user 2_Alex

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

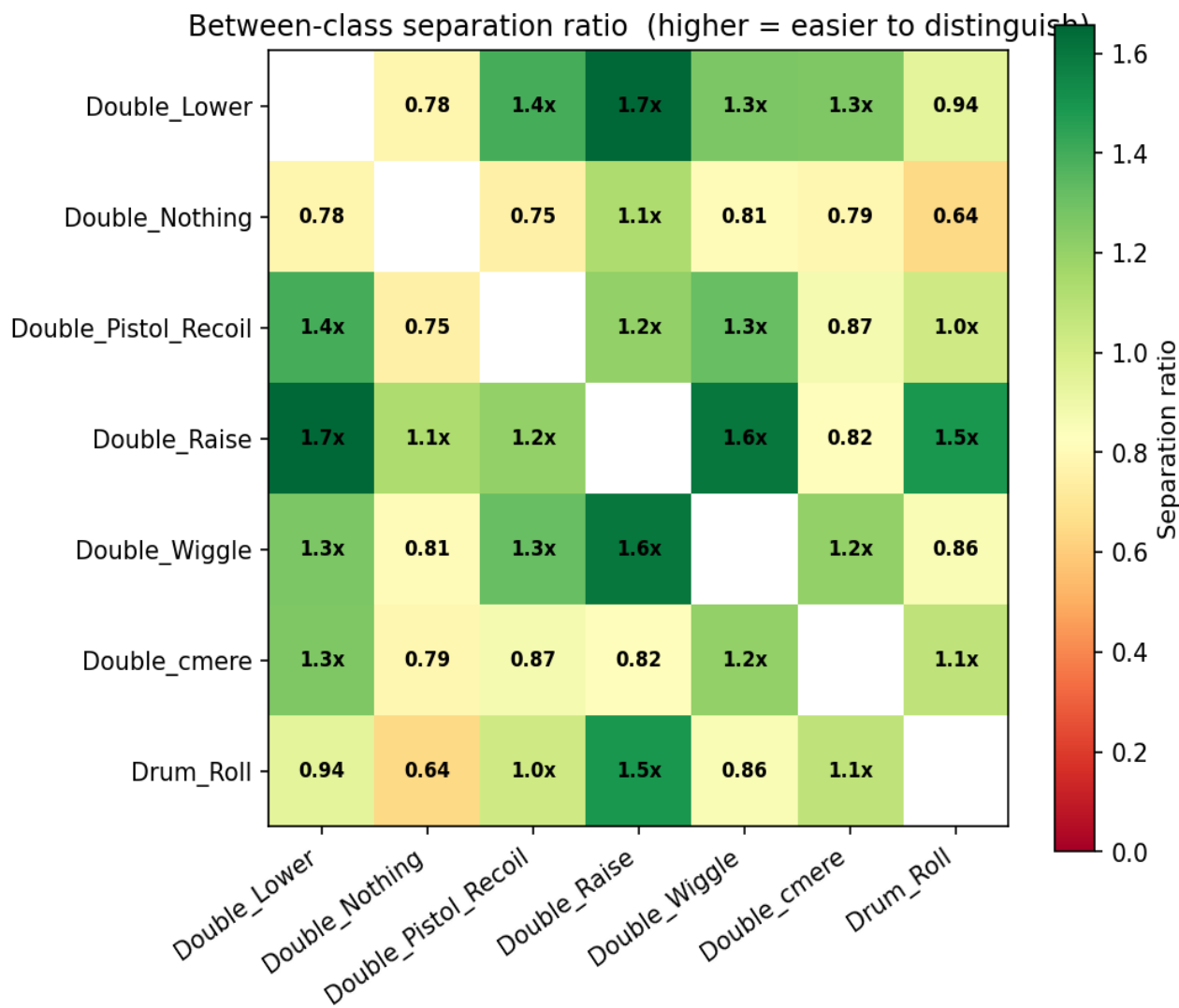
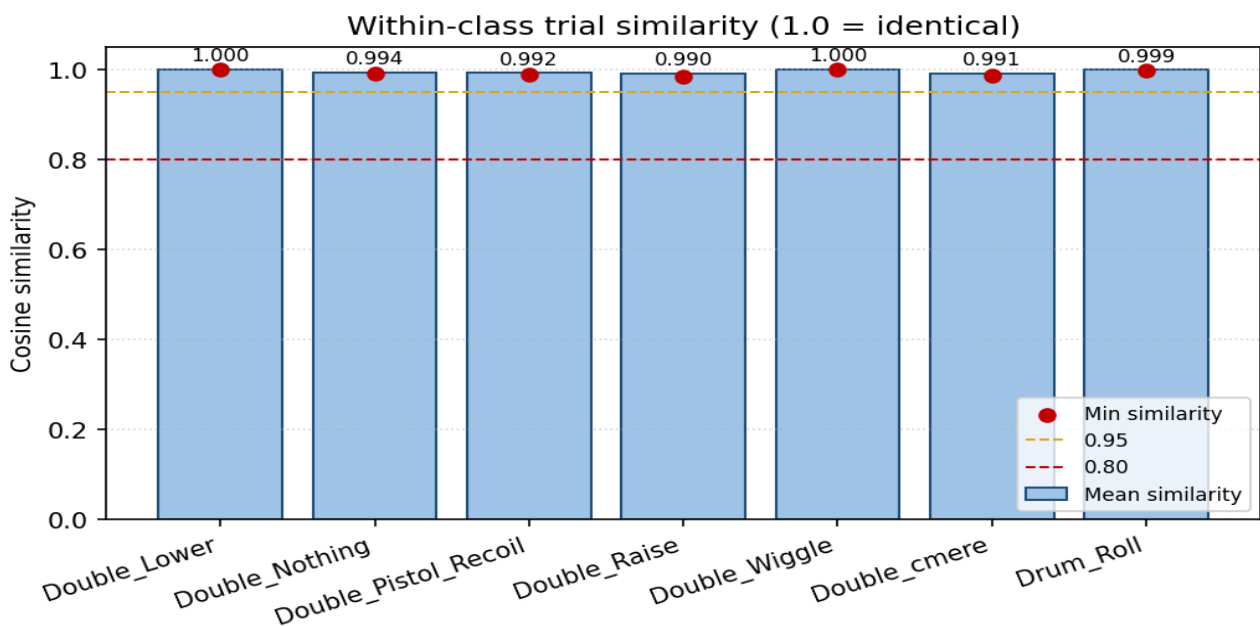


t-SNE feature space projection

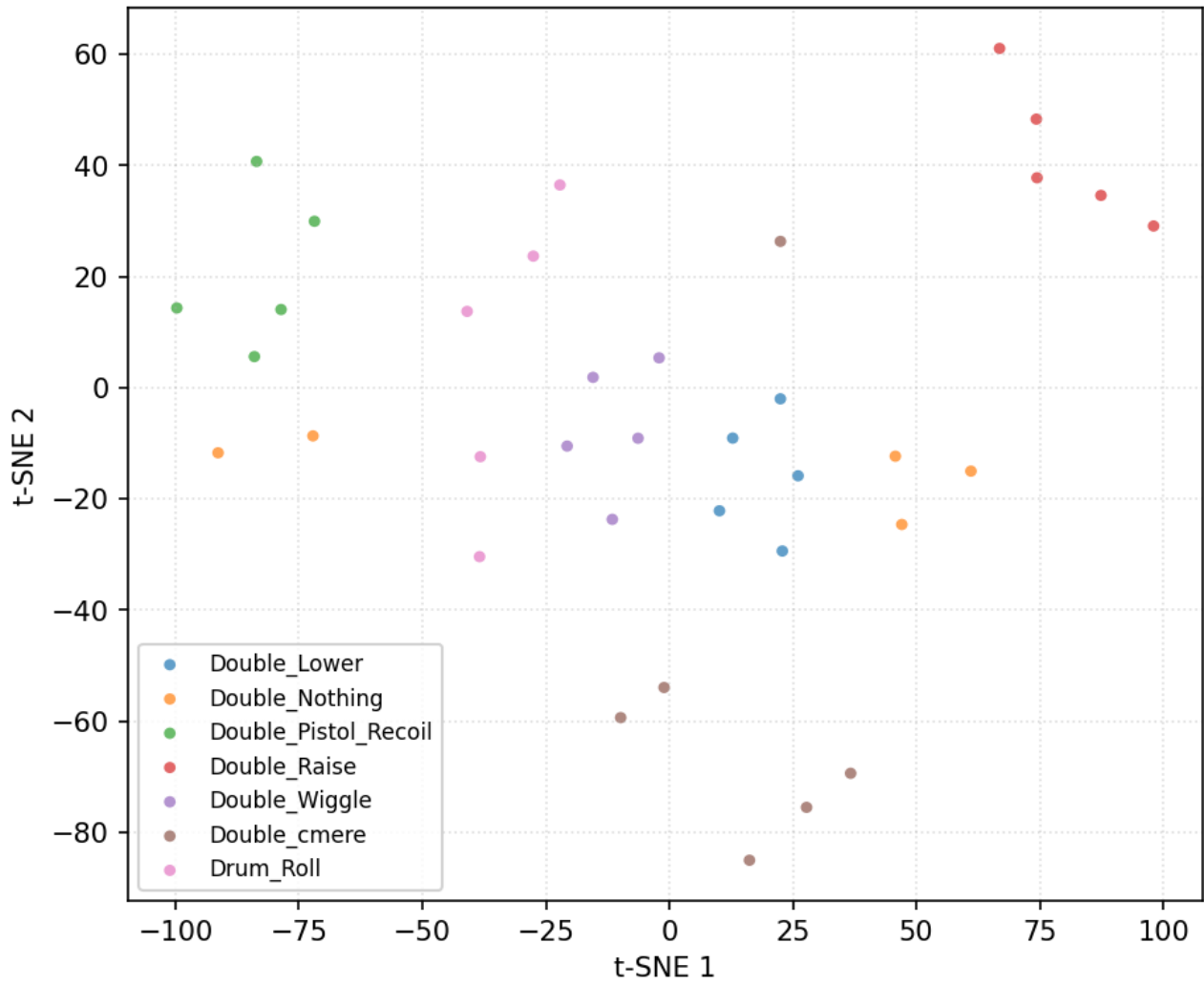


Data diagnostics — user 3_Anghad

36 trials, 7 classes (filtered + resampled, before per-trial normalisation).

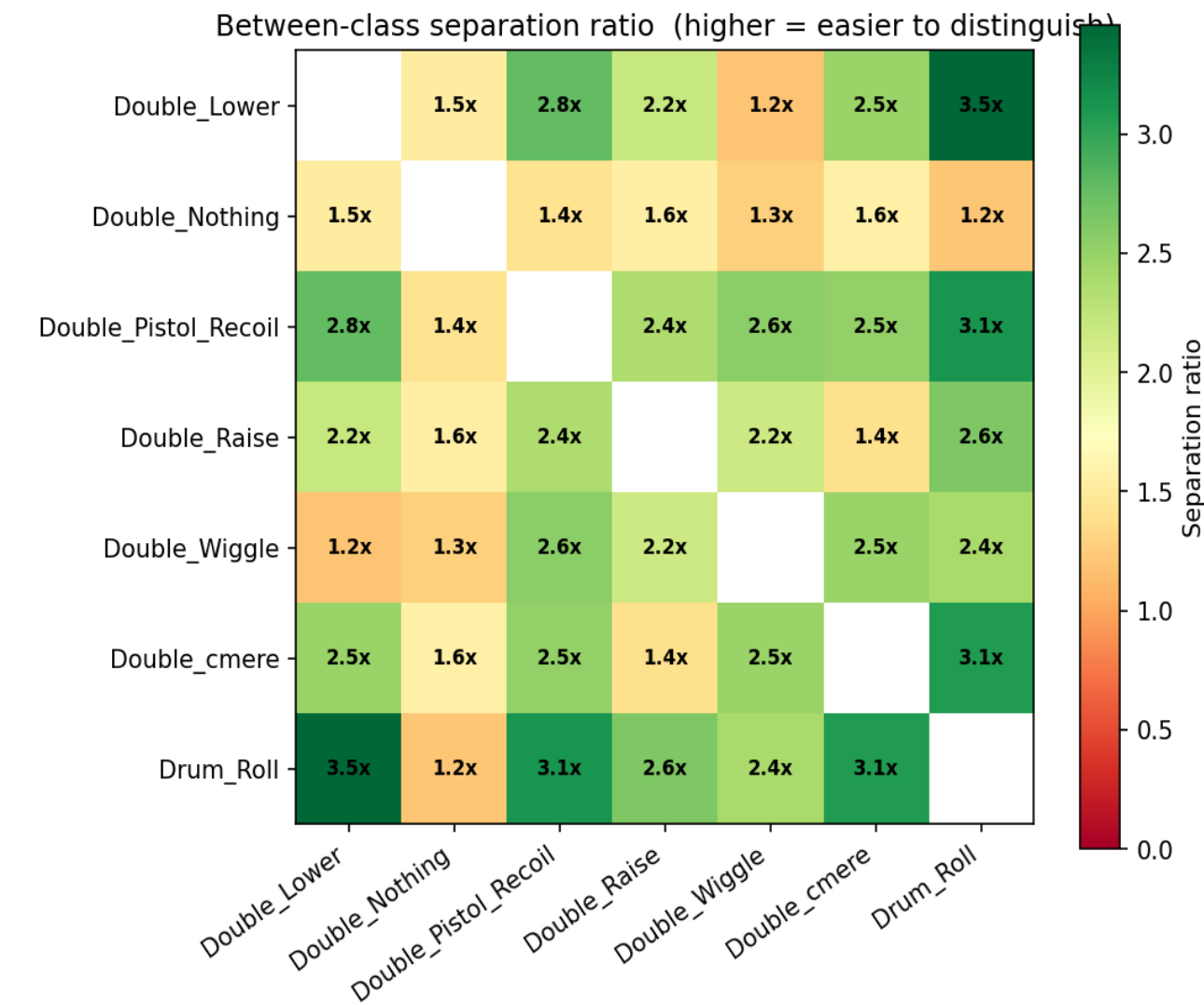
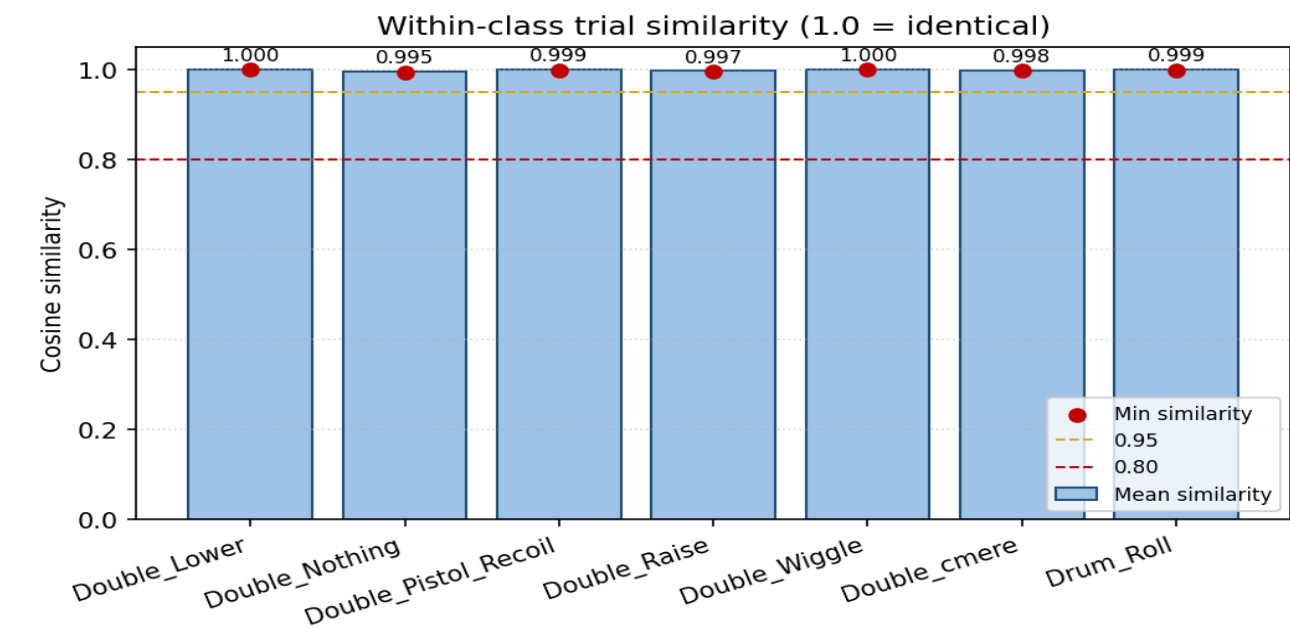


t-SNE feature space projection

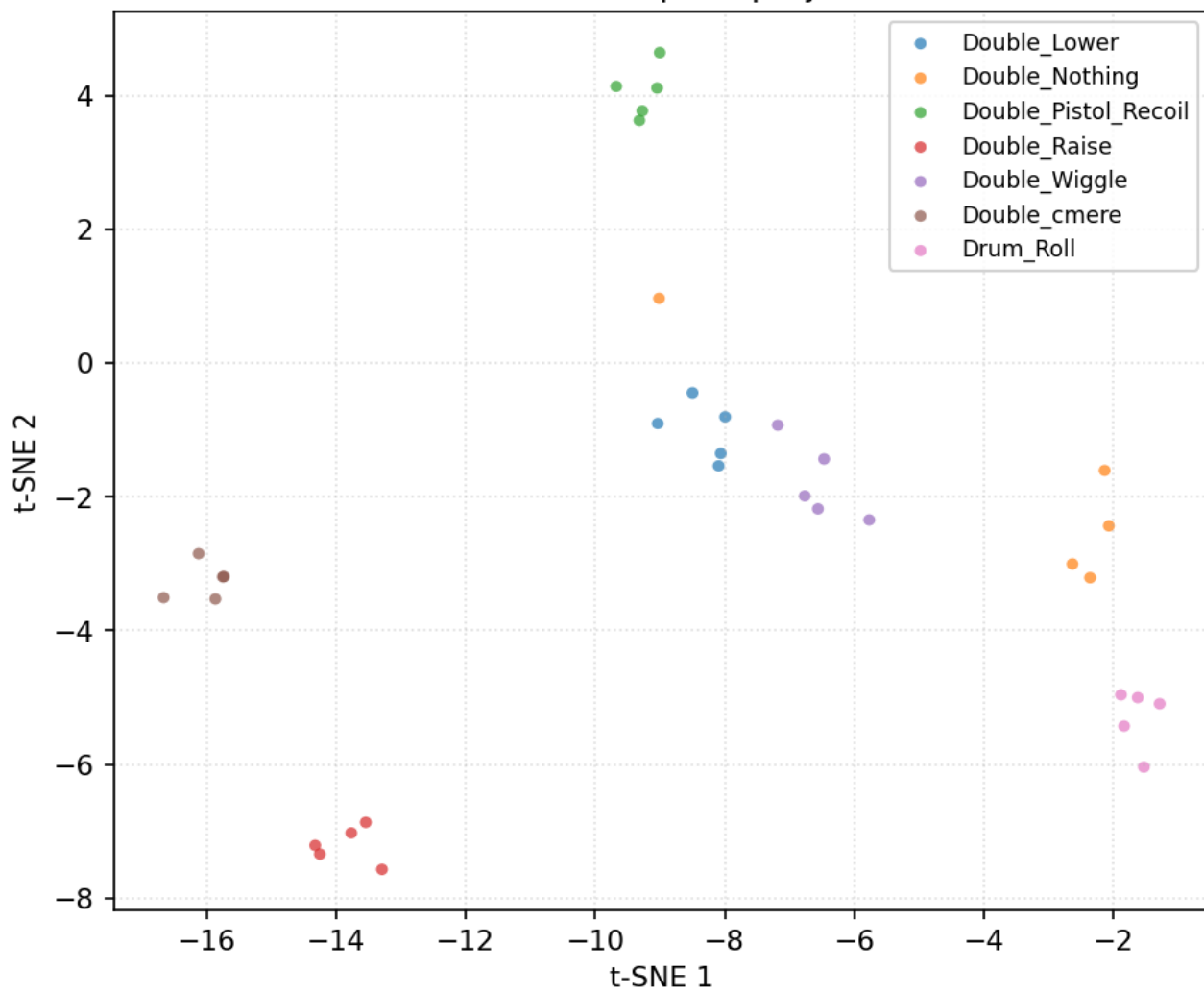


Data diagnostics — user 4_Daniel

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

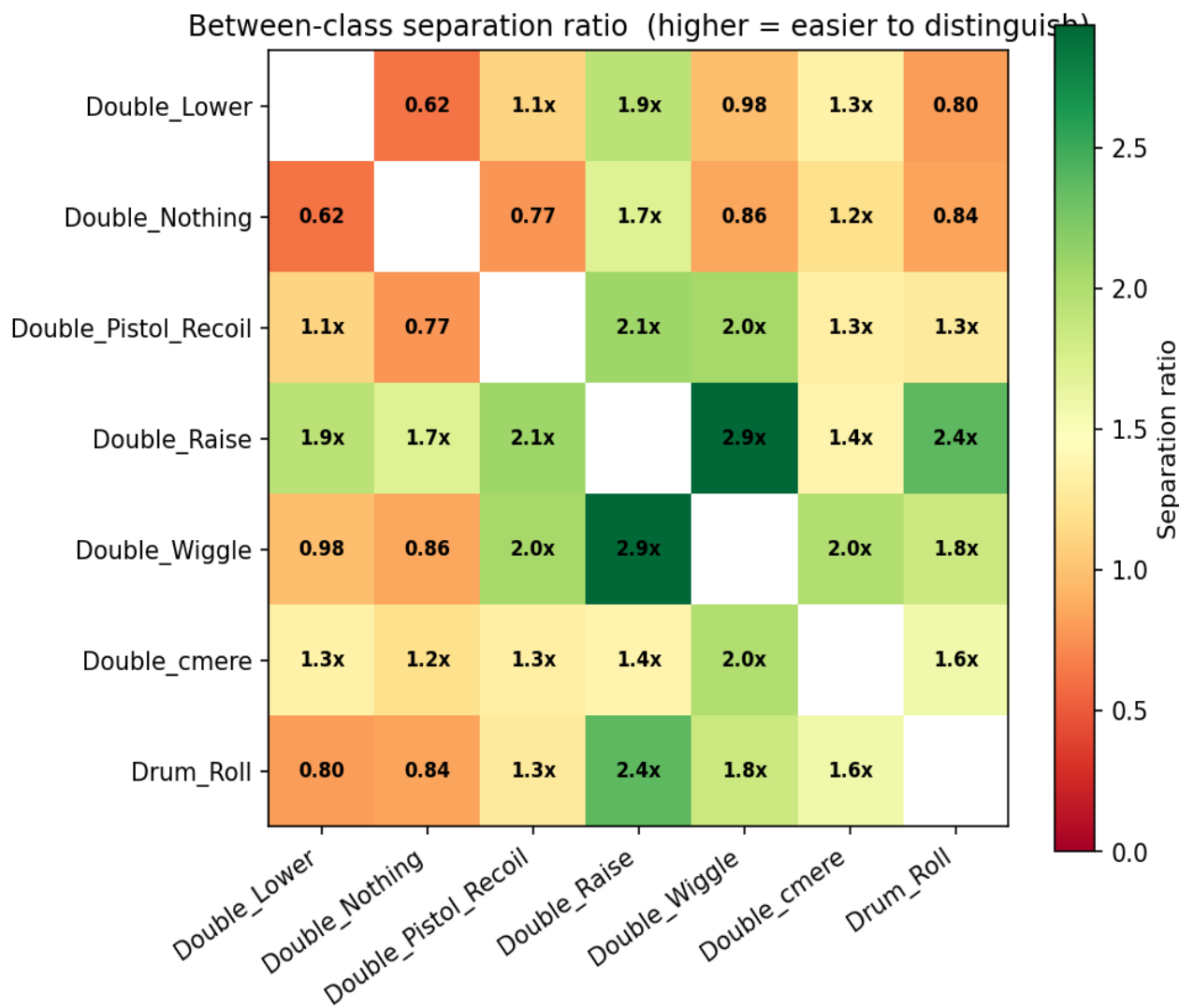
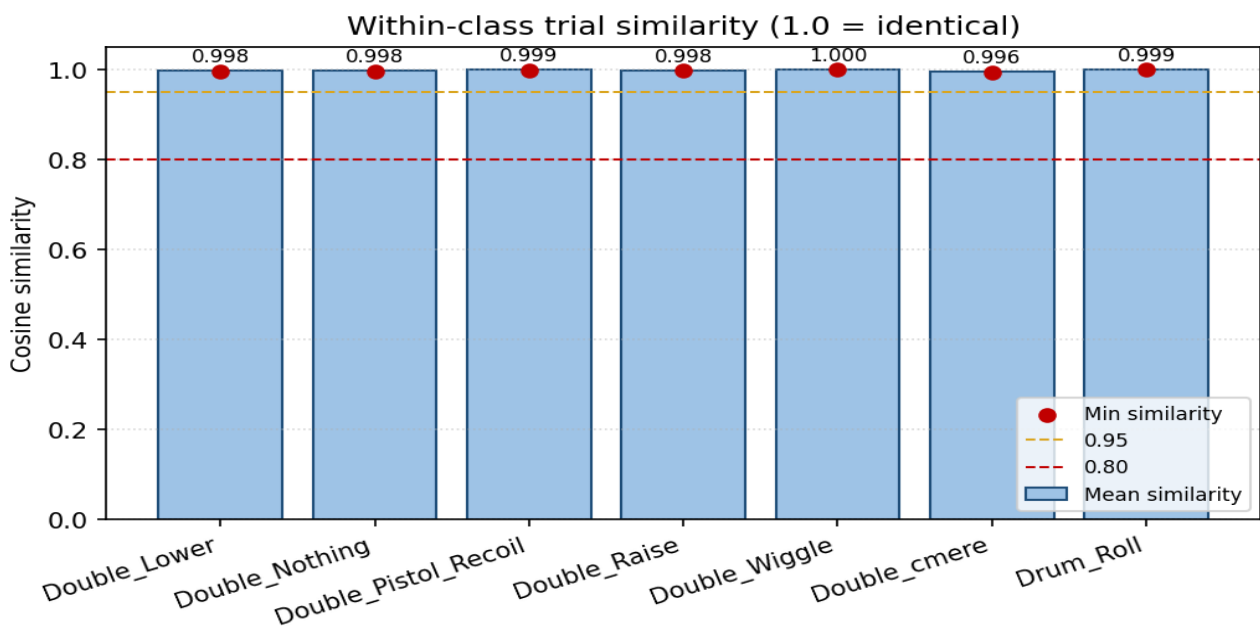


t-SNE feature space projection

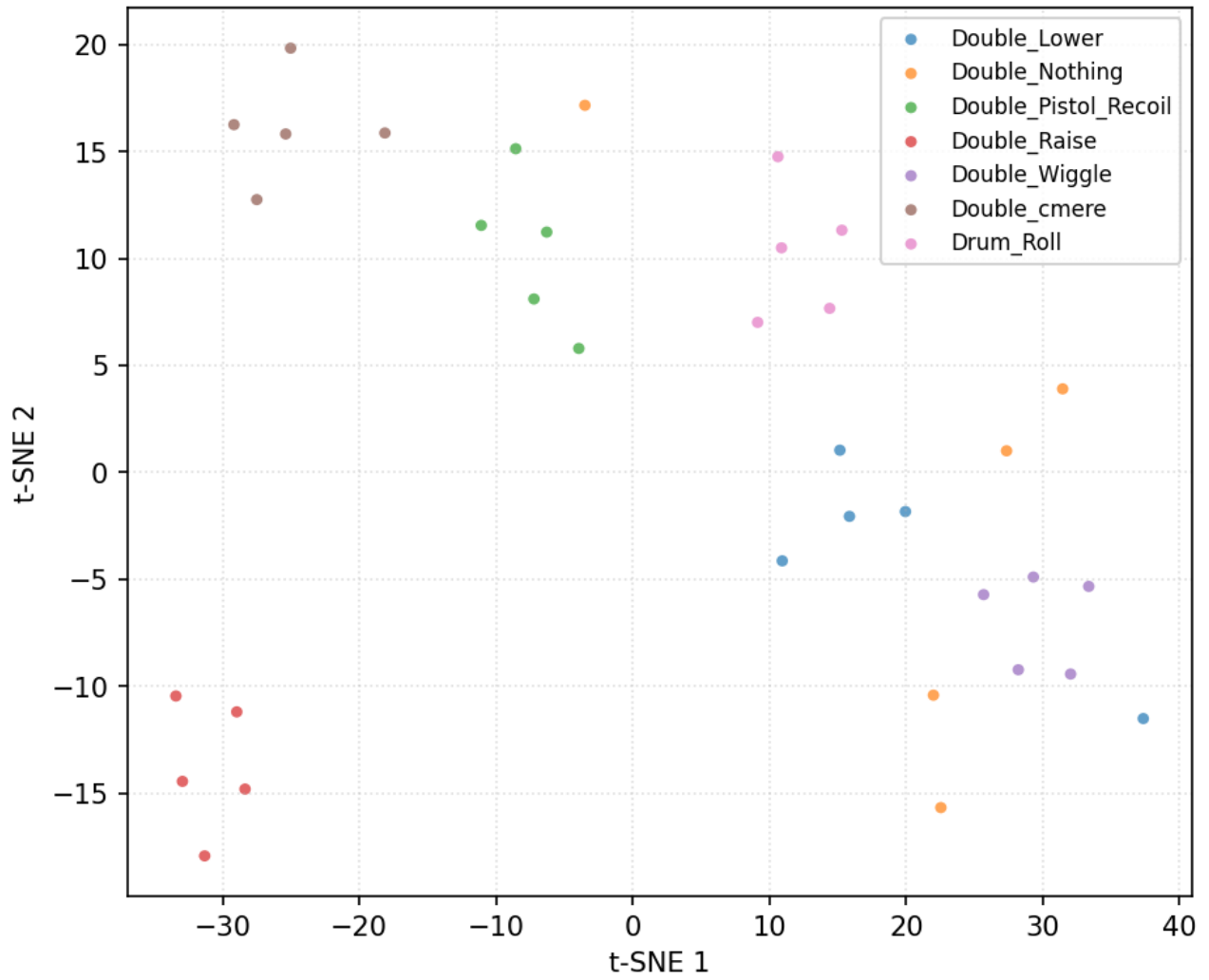


Data diagnostics — user 6_Harry

35 trials, 7 classes (filtered + resampled, before per-trial normalisation).

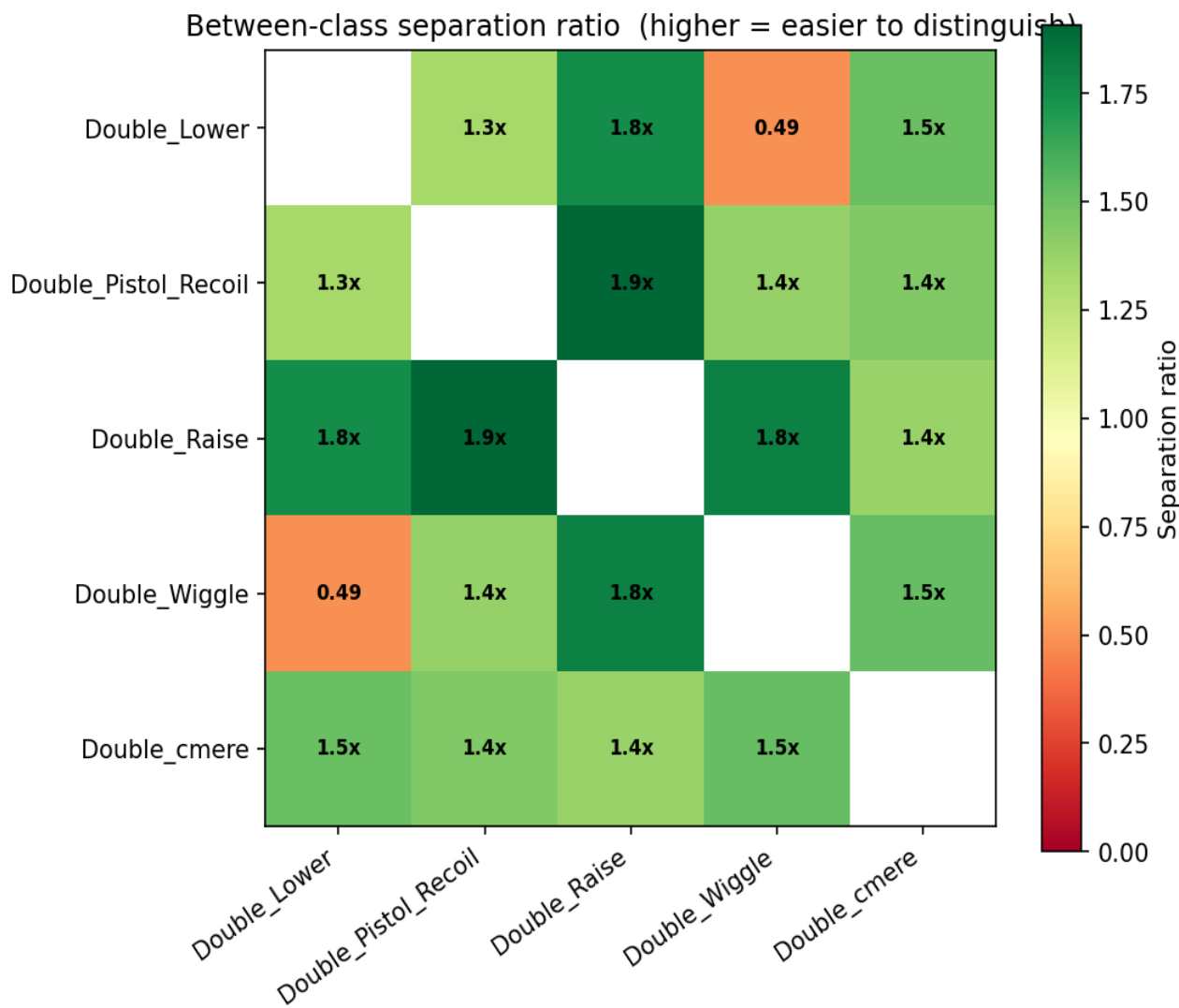
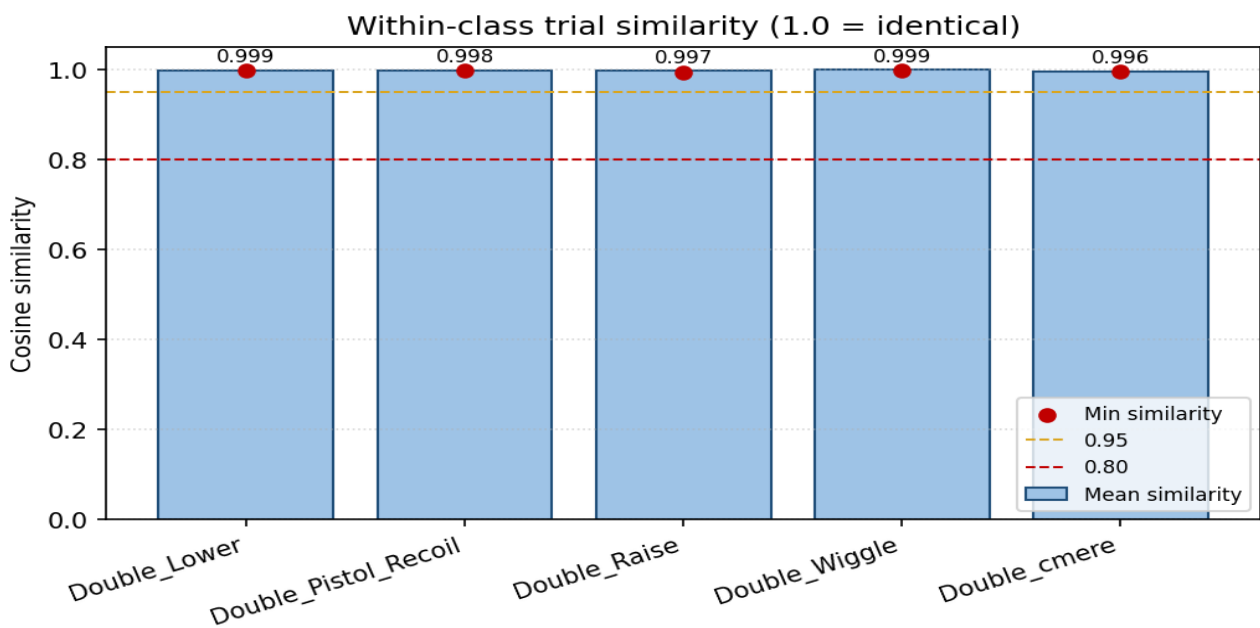


t-SNE feature space projection

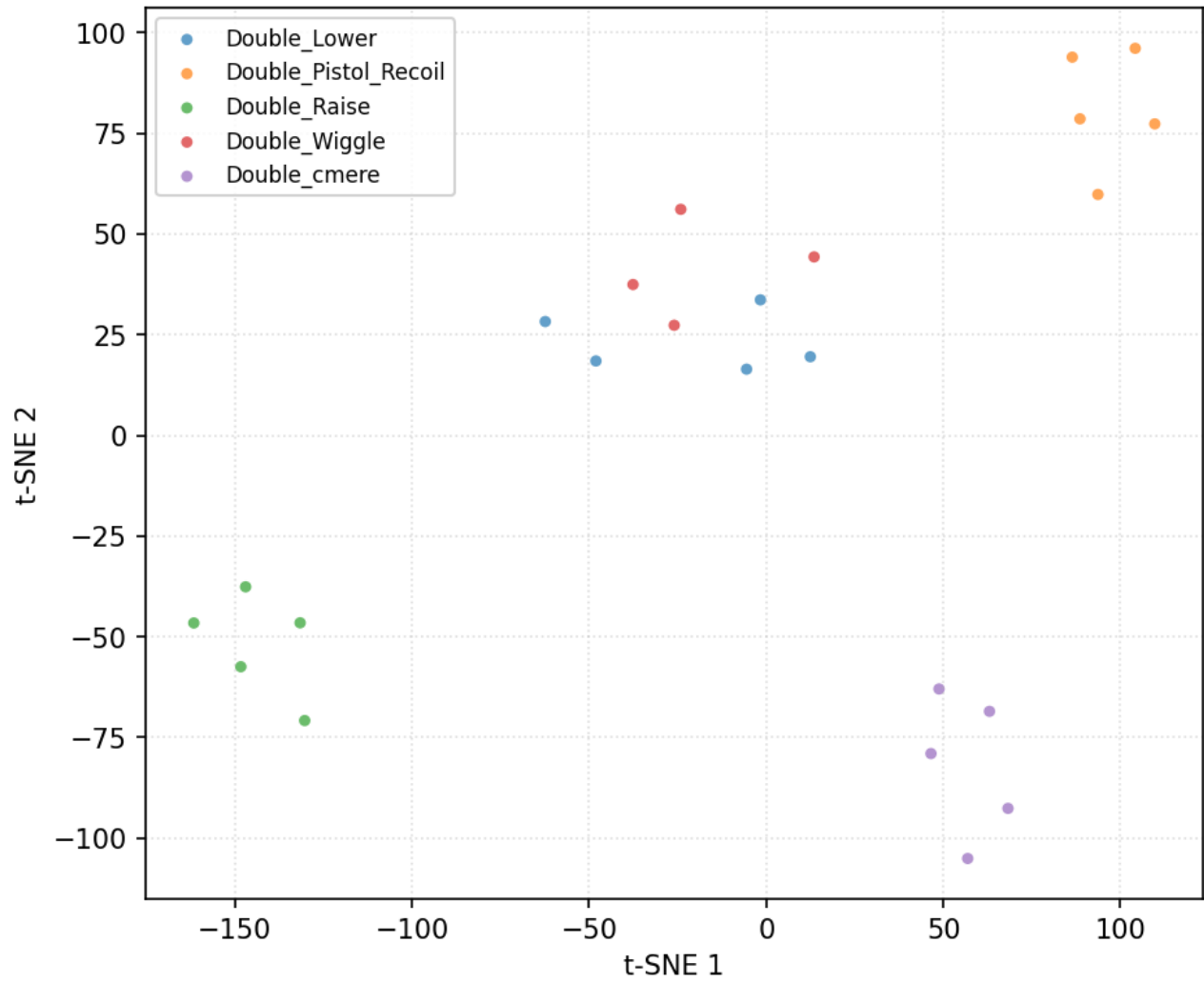


Data diagnostics — user 8_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



Variant: BN_GAP_Wide

Conv1D(32,k=8)+BN→Conv1D(64,k=8)+BN→Conv1D(128,k=5)+BN→GAP→Dropout(0.3)→Softmax [LOUO leader]

Trainable params	104,423
LOUO mean ± std	0.9436 ± 0.0656
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

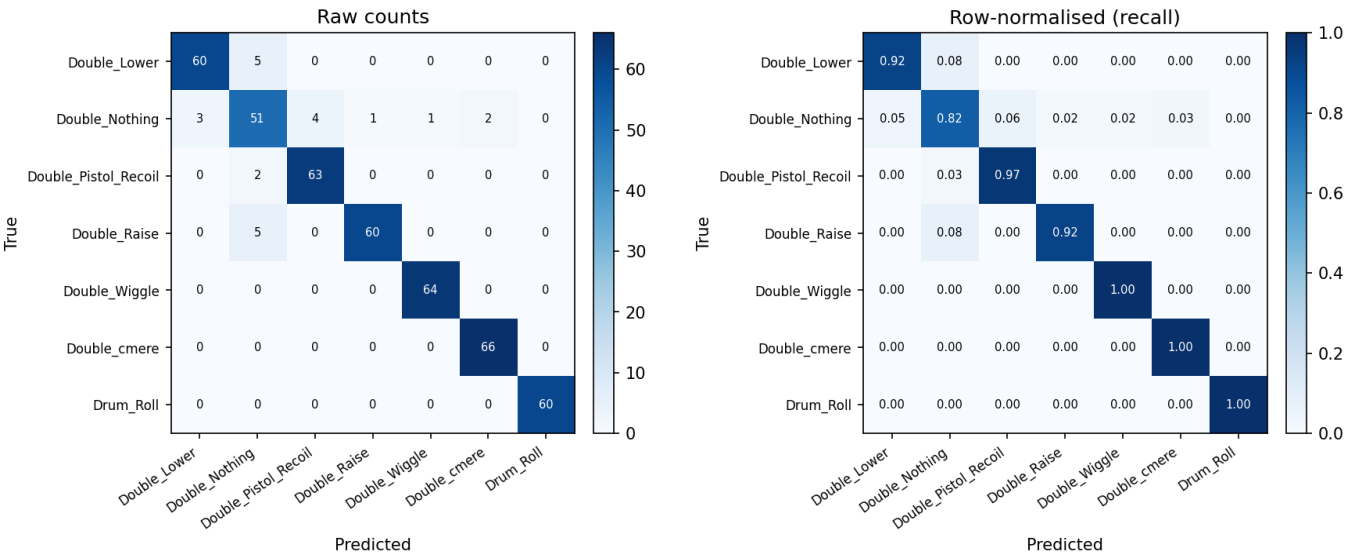
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	754	35	0.9143
2	12_Marcus	14_StephenV2	752	35	1.0000
3	14_StephenV2	15_Tash	752	36	0.9722
4	15_Tash	16_Bella	752	35	1.0000
5	16_Bella	17_Raquel	752	36	0.9444
6	17_Raquel	18_Bridgette	754	35	1.0000
7	18_Bridgette	1_Alan	754	35	0.9714
8	1_Alan	2_Alex	754	35	0.8857
9	2_Alex	3_Anghad	752	35	0.9429
10	3_Anghad	4_Daniel	752	36	1.0000
11	4_Daniel	6_Harry	754	35	0.9429
12	6_Harry	8_Jestin	776	35	0.9429
13	8_Jestin	11_Mansh	776	24	0.7500

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.952	0.923	0.938	65
Double_Nothing	0.810	0.823	0.816	62
Double_Pistol_Recoil	0.940	0.969	0.955	65
Double_Raise	0.984	0.923	0.952	65
Double_Wiggle	0.985	1.000	0.992	64
Double_cmere	0.971	1.000	0.985	66
Drum_Roll	1.000	1.000	1.000	60
macro avg	0.949	0.948	0.948	447
weighted avg	0.949	0.949	0.949	447

Confusion matrix (aggregated across folds)



Final retrain (deployment model)

Variant	Deep
Held-out user	— (trained on all users)
Held-out accuracy	—
Training-set size	894 samples (447 before augmentation)
Trainable params	204,007
Final train loss	0.0006
Final train accuracy	1.0000
Model path	— (not saved)
Labels path	— (not saved)
Scaler path	— (per-trial norm in use)